Highlights

Mapping the Causal Structure Between Home Energy Efficiency and Mental Health: A Directed Acyclic Graph Approach

Chun Sang Au Yong,Phil Symonds,Giorgos Petrou

- Energy efficiency and mental health links lack an explicit causal structure.
- We construct the first expert-reviewed DAG for home energy efficiency and mental health.
- 25 nodes and 89 edges separate confounders, mediators, and colliders at LSOA level.
- The backdoor criterion returns a five-variable minimal adjustment set.
- PC-stable and FCI corroborate the graph; efficiency and mental health are never adjacent.

Mapping the Causal Structure Between Home Energy Efficiency and Mental Health: A Directed Acyclic Graph Approach

Chun Sang Au Yong[a,*], Phil Symonds[a] and Giorgos Petrou[a]

[a] *UCL Institute for Environmental Design and Engineering, Central House 14 Upper Woburn Place, London, WC1H 0NN, United Kingdom*

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ABSTRACT

Residential energy efficiency is increasingly recognised as a structural determinant of population health, yet causal evidence linking home energy efficiency (HEE) to mental health remains limited. Observational studies in this field face high-dimensional confounding, do not distinguish confounders from mediators and colliders, and leave the pathways from energy efficiency to mental health unspecified. Estimates from such designs are liable to systematic bias and cannot support valid inference. This paper makes three contributions. First, we construct the first expert-informed directed acyclic graph (DAG) of the assumed causal structure between HEE and mental health, from a systematic synthesis of 53 peer-reviewed publications and independent expert review. The DAG comprises 25 nodes — exposure, outcome, confounders, mediators, and latent constructs — connected by 89 directed edges. Second, applying the backdoor criterion with the `dagitty` package in R, we derive the minimal sufficient adjustment set $\mathcal{A}$, comprising climate, economic capital, housing characteristics, tenure, and urbanicity. Third, we test the graph against constraint-based causal discovery, running PC-stable and FCI over the 16 constructs with an area-level proxy across 31,802 Lower Layer Super Output Areas (LSOA) in England. HEE and mental health are non-adjacent in the expert graph and in every discovered graph, at every significance level and under both outcome definitions. Encoding the expert causal ordering as background knowledge lowers the structural Hamming distance from 62 to 50 and raises v-structure recall from 0.01 to 0.14. FCI marks 28 of 39 edges as bidirected, including every HEE edge, indicating that area-level associations are dominated by unmeasured common causes — the premise on which the DAG's latent nodes and adjustment set rest. Discovery corroborates the graph; it does not estimate the effect. The DAG and its adjustment set establish the identification basis for the doubly-robust estimation reported in a companion paper.

1. Introduction

Buildings account for nearly 40% of global energy consumption and roughly one-third of greenhouse gas emissions (?). As nations accelerate decarbonisation to meet net-zero targets (?), residential retrofit has become a cornerstone of climate policy. The UK residential sector alone consumes 28% of national energy (?), and 19% of English households were in fuel poverty in 2022 (?). Improving home energy efficiency (HEE) therefore serves climate mitigation and fuel poverty objectives together. Its co-benefits for public health — particularly mental health — remain poorly quantified and, critically, poorly understood in causal terms.

The case for investigating these co-benefits is substantial. Mental disorders are one of the leading causes of disability worldwide, accounting for one in six years lived with disability (?). In England, excess cold in homes attributable to inadequate thermal efficiency costs the National Health Service an estimated £857 million annually (?). Cold and damp housing is associated with persistent worry, financial stress, depression, and anxiety (?), and studies across European contexts report a negative association between energy and fuel poverty and mental health (?). Intervention studies suggest energy efficiency improvements can improve mental wellbeing through enhanced thermal comfort, reduced financial burden, and psychosocial

*Corresponding author
thumbnails/cas_email.jpeg (C. Au Yong)
thumbnails/cas_url.jpeg (C. Au Yong)
ORCID(s): 0009-0009-5928-3301 (C. Au Yong); 0000-0002-6290-5417 (P. Symonds); 0000-0002-1524-2681 (G. Petrou)

empowerment (??). A causal interpretation nonetheless remains out of reach. As ? note, robust evidence on the causal relationships between home energy efficiency measures and mental health is lacking.

This gap reflects how existing observational studies have been designed rather than a shortage of data. Three limitations are salient. First, housing and health studies face high-dimensional confounding: a household's likelihood of receiving efficiency improvements is shaped by neighbourhood socioeconomic conditions (??), housing tenure (??), urbanicity (?), demographic composition (?), and local climate (?), all of which independently influence mental health (???). Without an explicit account of these pathways, no observed association can be attributed causally to the exposure. Second, standard regression does not distinguish confounders from mediators. The distinction matters because conditioning on a variable on the causal pathway induces bias rather than removing it (following the framework of ?). The literature has not mapped the mechanisms through which energy efficiency may affect mental health, leaving indirect pathways — thermal comfort, indoor air quality (?), financial stress relief (?), and psychosocial factors (?) — either over-adjusted or omitted. Third, prior adjustment strategies have been chosen on statistical association rather than causal reasoning, which systematic reviews of DAG use identify as a major source of residual confounding and unnecessary bias (?).

Directed acyclic graphs (DAGs) address these limitations directly. A DAG encodes explicit assumptions about the data-generating process as a directed graph; through d-separation and the backdoor criterion (?) it establishes whether a causal effect is identifiable from observational data and, if so, which variables must be conditioned on. The minimal adjustment set a DAG returns is not the largest available covariate set but the smallest sufficient to block all confounding backdoor paths without introducing bias through collider conditioning or mediator adjustment. Identifying and justifying that set is a prerequisite for any valid causal estimator applied to observational data (?).

This paper makes three contributions. First, we construct an expert-informed DAG of the assumed causal structure between home energy efficiency — operationalised through Energy Performance Certificate (EPC) ratings — and mental health, measured as depression prevalence and the Small Area Mental Health Index (SAMHI). The DAG integrates domain knowledge from the built environment, social epidemiology, and health geography literatures, and was revised through independent expert review. Second, applying the backdoor criterion with the `dagitty` package in R (?), we derive the minimal sufficient adjustment set required for unbiased estimation. Third, we confront the graph with the data it is intended to license: we test its implied conditional independencies directly, and run constraint-based causal discovery (PC-stable and FCI) over the constructs for which an area-level proxy exists. Discovery from cross-sectional area-level data cannot confirm a causal structure — it identifies an equivalence class at best — but it can contradict one, and we state before reporting results which disagreements would have required the graph to be redrawn. Together these contributions establish the identification basis for a companion paper that estimates the effect with doubly-robust causal machine learning.

Methodologically, the paper shows how DAG construction in a housing–health context requires systematic treatment of tenure-related selection (?), spatially structured confounding (??), and policy-induced heterogeneity (??). These challenges are endemic to built environment research but rarely addressed transparently within the causal inference literature.

2. Literature Review

2.1. Housing as a Determinant of Health

Housing is a well-established social determinant of health, acting through the physical quality of the dwelling, its thermal and acoustic environment, the financial stress of housing costs, tenure security, and neighbourhood characteristics (?). The Marmot Review placed housing conditions within a broader framework of structural health determinants, arguing that poor housing quality is among the most tractable material conditions generating health inequalities in England (?). Physical housing quality — crowding, dampness, thermal adequacy, structural integrity — predicts mental health deterioration in longitudinal studies (??). Using a prospective design, ? showed that objective measures of housing quality predict psychological distress independently of socioeconomic status, indicating that the dwelling environment influences mental health beyond the conditions determining access to it.

The mental health consequences of poor housing operate directly and indirectly. Directly, cold, damp, mould, and overcrowding generate chronic physiological and psychological stress (???). Indirectly, tenure insecurity and affordability pressures generate financial stress and erode perceived control, themselves risk factors for anxiety and depression (??). These pathways interact: ? document how energy poverty, food insecurity, and housing stress reinforce one another, with energy-poor households in Israel forgoing medication and healthcare to meet utility costs. This interlinking is why studies of any single housing attribute are prone to confounding bias whose direction and magnitude depend on the full causal structure rather than on the attribute studied in isolation.

2.2. Energy Efficiency, Fuel Poverty, and Mental Health

A specific strand of this literature treats home energy efficiency as a determinant of health. ?, in a realist review of 28 energy efficiency programmes, identify three channels: warmth (adequate thermal conditions reducing cold stress), affordability (reduced fuel costs alleviating financial anxiety), and psychosocial empowerment (improved home satisfaction, autonomy, and social status). The channels are not mutually exclusive, and their relative salience varies by household context, with financial stress relief most prominent among lower-income households (?).

Quantitative evidence spans several designs. Meta-analysing 36 studies of over 33,000 participants, ? found a small but significant positive effect of household energy efficiency interventions on health (weighted average effect size $d = 0.08$), larger among lower-income populations and for mental health outcomes specifically. ? confirmed that combined heating and insulation measures produced the most favourable effects on depression, anxiety, and social isolation. ? found significant improvements in mental health and psychosocial outcomes among residents of low-income areas in Wales following energy efficiency investment, with perceived warmth and financial wellbeing as key mediators. The randomised Victorian Healthy Homes Programme trial (?) — one of the few RCTs in this area — found that modest thermal upgrades improved mental health on the SF-36 mental component summary and reduced overall healthcare costs.

Fuel poverty has been linked to mental health in longitudinal analyses. Using Australian panel data, ? estimated that energy poverty increased the risk of mental health disorders, cardiovascular disease, and respiratory conditions, with mental health effects persisting after adjustment for socioeconomic confounders. A European scoping review identified 15 studies reporting associations between energy and fuel poverty and mental health, with women, children, and older adults disproportionately affected (?). In England, ? documented the persistent spatial concentration of fuel poverty over a decade, in patterns that map closely onto gradients in mental health service use and antidepressant prescribing (?).

The inferential status of this evidence remains associational. Most studies use cross-sectional or uncontrolled before-and-after designs that cannot address confounding (??). Longitudinal designs have generally focused on individual-level interventions without modelling the neighbourhood-level factors — deprivation, tenure composition, demographic profile, urbanicity — that shape the distribution of both energy efficiency and mental health (?). No study in this domain has formalised the assumed causal structure as a DAG, leaving the choice of adjustment variables implicit, inconsistent across studies, and unverifiable against any declared set of causal assumptions.

2.3. The Built Environment, Causal Inference, and the Challenge of Confounding

Causal inference in the built environment is difficult for structural reasons. Built environment exposures — housing quality, green space access, noise, air quality, neighbourhood deprivation — are not randomly distributed. They cluster with socioeconomic disadvantage, demographic composition, and tenure type, creating dense confounding structures (???). Any association between a built environment attribute and a health outcome is therefore susceptible to confounding not by a single covariate but by an interlocking network of socioeconomic, demographic, and geographic factors.

The green space and mental health literature is the closest methodological parallel. Cross-sectional studies consistently find inverse associations between residential greenness and depression and anxiety (??), but these are vulnerable to confounding by income, deprivation, and physical activity. Quasi-experimental and longitudinal approaches have begun to address this: ? used a ten-year dynamic panel study of 2.3 million adults in Wales to demonstrate greenspace–mental health associations robust to individual-level heterogeneity, and ? applied difference-in-differences and synthetic control methods to built environment

changes and subjective wellbeing in England. These designs rely on temporal variation in exposure rather than on modelling the confounding structure, so they cannot account for time-varying confounding or identify the pathways through which effects operate.

The DAG is the natural complement to these designs. Rather than relying on identifying variation, it makes the researcher's causal assumptions about the data-generating process explicit, enabling formal identification of the minimal adjustment set and flagging variables that should not be adjusted for because they are mediators or colliders. Reviewing 234 health research articles using DAGs, **?** found substantial variation in reporting: only 21% stated their target estimand explicitly and fewer than half reported the adjustment set their DAG implied. DAG-based approaches remain uncommon in the built environment field. One exception is **?**'s DAG for dwelling characteristics and household COVID-19 transmission, which identified housing tenure, overcrowding, and dwelling volume as requiring a priori adjustment — a structure that resonates with the confounding relevant here (**?**). More broadly, failing to distinguish confounders from mediators and colliders — endemic to covariate selection on statistical criteria alone — introduces bias as readily as under-adjustment (**??**).

2.4. UK Energy Efficiency Policy and the Distributional Context

The policy context is essential to the confounding structure the DAG must represent. Home energy efficiency in England is not randomly distributed: EPC ratings are systematically lower in older private rented properties, in fuel-poor rural areas, and in neighbourhoods with higher concentrations of deprivation (**??**). The private rented sector presents a structural barrier through the split incentive problem, in which landlords bear retrofit costs while tenants receive the bill savings, suppressing investment (**??**). Policy programmes including the Green Deal, the Energy Company Obligation, and the Warm Homes Discount have sought to address these inequities, but their reach has been geographically uneven and their take-up shaped by the same socioeconomic and tenure characteristics that influence mental health (**??**). Estimating the causal effect of energy efficiency on mental health therefore requires modelling the policy-driven selection mechanism that determines which neighbourhoods receive improvements, because that mechanism itself introduces confounding that covariate adjustment cannot address until it has been mapped causally.

3. Methods

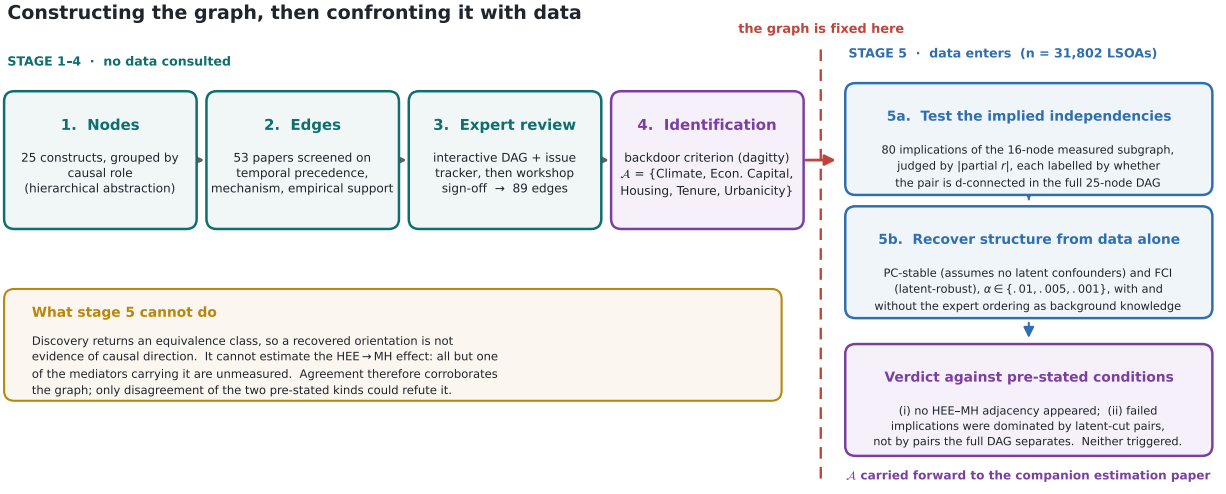


Figure 1: Construction and checking sequence. Stages 1–4 are knowledge-driven and consult no data; the graph is fixed before the data are opened. Stage 5 tests the conditional independencies the graph implies and compares it with structures recovered by constraint-based discovery. The checks can corroborate the graph but cannot establish edge directions or estimate the effect, for the reasons stated in the panel at lower left.

The approach proceeds in three stages. First, we specify the causal structure of interest as a directed acyclic graph, constructed from domain knowledge and systematic literature synthesis before, and independently of, any consideration of the data to be analysed. Second, we apply the backdoor criterion to the DAG using the `dagitty` package in R (**?**) to derive the minimal sufficient adjustment set. Third, we confront the resulting graph with area-level data, testing its implied conditional independencies and comparing it against structures recovered by constraint-based causal discovery. The ordering is deliberate and follows the principle that structural assumptions must precede estimation (**??**): a DAG encodes a theory of the data-generating process, and its construction should reflect domain knowledge rather than the statistical properties of any particular dataset.

This contrasts with data-driven variable selection — stepwise regression, penalised regression, feature importance ranking — which selects covariates by their empirical association with outcome or exposure. Such strategies produce adjustment sets that conflate confounders, mediators, and colliders, so conditioning on the resulting set may introduce bias even where it reduces variance (**?**). Grounding covariate selection in an explicit causal graph ensures the adjustment set reflects the theoretical confounding structure rather than the idiosyncrasies of the sample. The third stage does not reverse this ordering: discovery is run after the graph is fixed, and is used to test the graph, not to build it. Figure 1 sets out the sequence in full, including the point at which the data are first consulted.

3.1. Node Specification: Conceptual Grouping and Variable Abstraction

DAG construction begins with the nodes — the variables whose causal relationships the graph encodes. The relevant domain here spans housing characteristics, energy systems, socioeconomic conditions, demographic composition, environmental exposures, neighbourhood-level social structures, and health outcomes. We organise these into 25 theoretically coherent nodes by hierarchical abstraction, grouping constructs that share a common causal role relative to the exposure and outcome into a single node.

This abstraction is a theoretical operation, not a data reduction step. The 25-node structure follows from reasoning about what constitutes a distinct causal entity in the housing–health system: what constructs have independent causal relevance that merging would obscure, and what separations would add graph complexity without adding causal content. Each node's hypothesised role — confounder, mediator, exogenous variable, or latent root cause — is specified a priori on substantive grounds and treated as a prior to be interrogated, not a conclusion. Table 1 gives the full node schema with a brief description of each construct and its hypothesised role.

Table 1: Node schema for the causal DAG: 25 constructs with hypothesised causal roles. Roles represent theoretical priors, specified prior to formal identification analysis and subject to revision through expert review.

Node	Description	Hypothesised Role
Exposure and Outcome		
HEE	Home Energy Efficiency: insulation, glazing, heating system quality, and airtightness	Exposure
MH	Mental Health: rates of depression, anxiety, chronic stress, and related outcomes	Outcome
Structural Confounders		
Economic_Capital	Financial resources including income levels and employment status	Primary confounder
Education	Educational attainment and health literacy	Primary confounder
Housing_Characteristics	Physical dwelling form: building age, construction era and dwelling type (detached, semi-detached, terraced, flat)	Physical confounder

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Table 1 continued.

Node	Description	Hypothesised Role
Tenure	Legal occupancy status: owner-occupied, private rental, or social rental	Structural confounder
Demographics	Age, race, gender, and family structure	Baseline confounder
Urbanicity	Degree of urban versus rural setting	Macro-confounder
Climate	Exogenous climate conditions: regional temperatures, flood risk, and extreme weather	Exogenous variable
Proximal and Environmental Mediators		
IEQ	Indoor Environmental Quality: thermal comfort, damp, mould, and indoor pollutants	Proximal mediator
Fuel_Poverty	Household inability to afford adequate warmth, operationalised via the Low Income Low Energy Efficiency (LILEE) indicator; jointly determined by dwelling efficiency, energy prices, and household income	Proximal mediator
Air_Pollution	Ambient outdoor air quality ($PM_{2.5}$, NO_2)	Environmental mediator
Green_Space	Access to and quality of parks, trees, and blue spaces	Environmental mediator
Noise	Area-level environmental noise (road, rail, aircraft) measured via L_{den}/L_{night}	Environmental mediator
Overcrowding	Density of residents relative to number of rooms	Physical mediator
Social and Structural Mediators		
Neighbourhood	Latent area-level construct representing the geographic unit at which household-level factors aggregate into shared environmental exposures; the node through which household-level and structural causes are transmitted to area-level environmental and social mediators	Scale-bridging construct
Neighbourhood_Stress	Chronic neighbourhood-level stressors, including crime, perceived unsafety, physical disorder and dereliction, and anti-social behaviour	Social mediator
Accessibility	Physical ease of access to healthcare, employment, and services	Social mediator
Social_Capital	Area-level civic participation, community trust, and associational density	Social mediator
Digital_Capital	Broadband availability, speed, and digital literacy or exclusion at the area level	Structural mediator
Policy_Scheme_Access	Area-level uptake of, and eligibility for, energy assistance and weatherisation programmes (e.g. ECO, Warm Home Discount)	Policy mediator
Isolation	Social isolation, loneliness, and lack of community belonging	Psychological mediator
Health Mediators		

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Table 1 continued.

Node	Description	Hypothesised Role
PH	Physical Health: respiratory health, cardiovascular disease, and mortality	Health mediator
Substance_Abuse	Area-level prevalence of alcohol and drug misuse	Health mediator
Latent Constructs		
U_AreaHistory	Unobserved legacy of historical planning decisions, disinvestment, and structural disadvantage	Unmeasured root cause

3.2. Edge Specification via Systematic Literature Synthesis

With nodes defined, we specified the directed edges — the assumed causal relationships — through systematic review of the empirical and theoretical literature. Inclusion required three conditions. First, *temporal precedence*: X must plausibly precede Y in the causal timeline, assessed on substantive grounds given the cross-sectional nature of many area-level constructs. Second, *mechanistic plausibility*: a credible physiological, economic, behavioural, or social pathway must connect X to Y; speculative associations with no proposed mechanism were excluded. Third, *empirical support*: at least one peer-reviewed study must document an association between X and Y after adjustment for major third-variable influences.

These criteria follow established guidance for evidence-based DAG construction in health research (**?**). The supporting literature for each included edge is documented and discussed in the Results alongside the DAG itself, where the causal structure is presented and justified in full. This sequencing reflects the epistemic status of the edges: they are hypotheses grounded in prior knowledge, justified substantively rather than procedurally.

We reviewed 53 peer-reviewed publications spanning environmental epidemiology, housing studies, social epidemiology, and mental health research. Candidate edges were enumerated by the research team, screened against the three conditions, and carried forward to expert review.

3.3. Expert Review and Structural Refinement

To guard against specification errors arising from any single research team's blind spots, the provisional DAG was submitted to structured expert review. The graph was published as an interactive visualisation in which each node and edge can be inspected together with its stated rationale and supporting references.[1] Reviewers could act in two ways. They could react to existing proposals — each proposal being a suggested addition, removal, or reversal of an edge — which the visualisation aggregates as a running tally; or they could file a new proposal against a specific node or edge, giving a justification, a mechanism, and a citation where available. Proposals could be submitted either through a public issue tracker or as an annotated PDF returned by email, consolidated into a single record; the second channel was provided so that reviewers unwilling to create an account were not excluded. All submissions were visible to all reviewers, who were invited to respond to one another, so the review was deliberative rather than a set of independent parallel assessments. The tracker preserves the audit trail in a form that can be reconstructed after publication.

Eight reviewers external to the project team were invited, drawn from five institutions in two countries. Their fields span building physics and housing-stock modelling, environmental epidemiology, indoor environmental quality and child health, health psychology, behavioural economics, and building energy data. Reviewers were asked three questions: whether any important causal pathway was missing, whether any edge was implausible or insufficiently supported, and whether the hypothesised role of each node was appropriate given their domain knowledge.

Review proceeded in three stages. In the first, the graph was opened for asynchronous comment over a three-week window, and proposals were answered as they arrived so that the workshop could be reserved for what remained contested. The second stage was a ninety-minute workshop, recorded and transcribed,

[1] https://csauyong.github.io/dag-review/

structured as a shared reading of the whole graph and of the identifying assumptions followed by a sector-by-sector review of five groups of nodes. The third stage circulated the revised graph to all invitees, including those unable to attend, left the written channels open, and requested sign-off before estimation proceeded.

The two channels produced complementary yields. The asynchronous stage received ten substantive proposals from three reviewers, flagging six of the 25 constructs, and concentrated on the scope and definition of individual nodes; eight arrived before the workshop and two during the sign-off round. The workshop produced 26 substantive interventions from the five reviewers who took part — four of those invited, together with one further colleague — and concentrated instead on the structure and on the identifying assumptions. Nine of these resulted in a change to the graph, to a node definition, or to the analysis plan. Reporting the two counts separately reflects a division of labour that was intended: written review surfaced definitional problems in advance, and the session was spent on structure.

Table 2 records the resolution of each written proposal, and Table 3 records each change arising from the workshop together with the concern that prompted it. Every proposal received before the sign-off round was resolved as accepted, accepted with modification, or declined, with a stated rationale; two proposals received during the sign-off round remain open at the time of writing and are shown as such.

One limitation of the instrument itself should be recorded. The per-node construct definitions were held in the project's node schema but were not exposed in the interactive visualisation during the first two stages, an implementation omission identified during the workshop. This accounts for the concentration of asynchronous comments on node scope and definition. A definitions view was added to the visualisation before the sign-off round, which was therefore conducted with the definitions available.

Table 2: Resolution of the ten substantive proposals received through the public issue tracker. Numbers are the tracker's own issue numbers; two further issues are the authors' tests of the submission template and are excluded. Proposals 3–10 were received before the workshop and answered the day before it; 11 and 12 were received during the sign-off round.

Target and proposal	**Decision**	**Rationale and outcome**
(3) *Green_Space* should capture views of green space, not only proximity	Accepted into the edge rationale	The mechanism was added to the *Green_Space* → MH rationale. No structural change: the proposal concerns what the construct denotes, not which edges it carries.
(4) *Overcrowding* should extend to adjacency and residential density in multi-dwelling buildings	Declined, with re-routing	*Overcrowding* is retained as within-dwelling occupancy. The adjacency mechanism is built form and residential density, which the graph routes through *Housing_Characteristics* (dwelling type) and *Noise*; that route was made explicit rather than a density node added.
(5) Fuel prices are missing as a cause of *Fuel_Poverty*	Accepted causally, declined structurally	Price is a genuine cause of fuel poverty alongside income and dwelling efficiency. It is, however, a national and time-common input, so it cannot lie on a backdoor path between areas at a given time and is absorbed by the between-area contrast.

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Table 2 continued.

Target and proposal	Decision	Rationale and outcome
(6) *Housing_ Characteristics* should be reclassified as a social mediator, since visible disrepair carries psychological impact	Declined; treated as support for an existing edge	A node's role follows from its edges relative to the exposure–outcome pair rather than from a label. *Housing_ Characteristics* has edges into both HEE and MH, which is why the backdoor criterion places it in $\mathcal{A}$. The disrepair mechanism is evidence for the *Housing_ Characteristics* $\rightarrow$ MH edge, one of the two that make it a common cause.
(7) *IEQ* should connect to *Fuel_ Poverty* through damp and mould	Accepted, with the direction reversed	Underheating driven by unaffordable energy lowers internal surface temperatures and promotes condensation, damp and mould, so the edge runs *Fuel_ Poverty* $\rightarrow$ *IEQ*. This creates a second route, HEE $\rightarrow$ *Fuel_ Poverty* $\rightarrow$ *IEQ* $\rightarrow$ MH, alongside the direct HEE $\rightarrow$ *IEQ* edge.
(8) *Housing_ Characteristics* should include maintenance and upkeep	Declined, with re-routing	Maintenance sits downstream of the structural node: it is shaped by *Tenure*, through the landlord–tenant split incentive, and by *Economic_ Capital*, and reaches mental health through dwelling condition and *IEQ*. It is therefore captured by existing routes rather than as a structural attribute.
(9) *Housing_ Characteristics* should include structural and safety defects, such as historic cladding	Accepted conceptually; not operationalisable	The mechanism supports a direct *Housing_ Characteristics* $\rightarrow$ MH channel through safety and security, distinct from the efficiency channel. Structural-safety status is not observable in the area-level property data used here, so it enters as an acknowledged unmeasured component of the construct.
(10) MH: the graph has no temporal dimension; and co-resident mental health, diet, genetic predisposition and the wellbeing/diagnosis distinction are absent	Mixed	The temporal point was carried into the workshop as an explicit assumption and is discussed in Section 5.2. Co-resident mental health, diet and genetic predisposition are individual-level constructs with no area-level operationalisation. The wellbeing/diagnosis distinction is partly addressed by retaining the SAMHI as a second outcome definition.

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Table 2 continued.

Target and proposal	Decision	Rationale and outcome
(11) *Housing_ Characteristics* are pre-existing structural attributes rather than mediators of current climate effects on *IEQ*	Open at the time of writing	Received during the sign-off round. The proposal bears on the *Climate* → *Housing_ Characteristics* edge.
(12) Add *Urbanicity* → *Accessibility* → MH, and *Neighbourhood* → *Accessibility*	Open at the time of writing	Received during the sign-off round. The second pathway is already present in the graph; the proposal is effectively for the direct *Urbanicity* → *Accessibility* edge.

Two decisions ordinarily taken by the analysts were put to the panel and settled there. The first is the operationalisation of the outcome: recorded depression prevalence and the SAMHI were presented with their trade-offs — a single clearly defined recorded condition whose change score avoids the pandemic window, against a broader composite whose change score spans it — and the panel was asked which should be primary and which should serve as a sensitivity analysis. The division adopted in this paper and its companion follows that discussion. The second is the set of identifying assumptions: seven were presented in order from definitional to contestable, and the two the panel judged load-bearing — that what stays fixed cancels out under differencing, and that no important confounder is missing — are the two that the companion paper subjects to explicit test.

Several concerns raised at the workshop were recorded without being adopted, either because no area-level operationalisation exists or because they bear on estimation rather than structure. These are set out in Section 5.2 rather than here, so that the limitations they imply are stated where a reader will look for them.

The principal structural change arising from review was the addition of *Fuel_ Poverty* as a second proximal mediator, carrying the affordability channel that the provisional graph routed entirely through indoor environmental quality (Section 4.1.4). The minimal adjustment set was re-derived after every structural revision.

3.4. Formal Identification of the Minimal Adjustment Set

With the DAG finalised, we applied the backdoor criterion (**?**) to identify the minimal sufficient adjustment set for the total causal effect of HEE on MH, using the `dagitty` package in R (**?**). The package enumerates all backdoor paths between exposure and outcome and returns the smallest variable set whose conditioning blocks them without conditioning on any mediator or collider.

The returned set $\mathcal{A}$ is minimal in the sense that no proper subset closes all backdoor paths and no member is redundant given the others. Variables absent from it despite correlating with both HEE and MH are excluded because they lie on or downstream of the causal pathway, where conditioning would induce post-treatment bias. The derivation and interpretation of $\mathcal{A}$ are presented as a primary result in Section 4.

3.5. Causal-Discovery Checks

The DAG is an assumption, and an assumption stated this precisely can be confronted with data. The final stage does so in two ways: by testing the conditional independencies the graph implies, and by comparing the graph against structures recovered from the data by constraint-based causal discovery. Both are corroboration exercises. Cross-sectional area-level data cannot establish causal direction, and discovery algorithms return equivalence classes rather than unique graphs, so agreement cannot confirm the DAG. Disagreement of the right kind can nonetheless contradict it, which is what makes the exercise informative.

3.5.1. Data and construct mapping

The checks use the same area-level dataset as the companion estimation paper: 31,802 LSOAs in England, after listwise deletion of records with missing or non-finite values, from a total of 32,844. Sixteen of the 25

Table 3
Changes arising from the workshop, and the concern that prompted each. Edge changes are structural; definition and rationale changes alter what a construct denotes or why an edge is drawn without altering the topology.

Change	Type	Concern that prompted it
Fuel_Poverty → *IEQ*	Edge added	Raised independently in both channels: in writing as a damp-and-mould link between the two constructs, and at the workshop as the observation that warmer homes have less damp, so the affordability channel may reach mental health indirectly as well as directly.
Accessibility → *PH*	Edge added	Distance and travel time to services delay diagnosis and disrupt the management of physical illness; the edge also opens an indirect *Accessibility* → *PH* → MH path.
Air_Pollution → *IEQ*	Confirmed	Queried for consistency, on the grounds that external noise is drawn as a source of indoor conditions and external air pollution arguably should be too. The edge was already present and is retained unchanged.
Climate → HEE	Removed, then reinstated	Challenged on the grounds that climatic variation within Great Britain is too narrow to drive retrofit decisions, and removed. Reinstated on subsequent review: absolute savings from a given fabric measure scale with underlying heat demand, which alters cost-effectiveness and the geographic targeting of retrofit, and therefore the efficiency state ultimately achieved. The edge is a member of the adjustment set, so the reversal is reported here rather than absorbed silently.
Accessibility redefined	Definition	The construct was read as physical distance alone; it was broadened to include the local availability of mental-health service provision, which shapes help-seeking behaviour and is unevenly distributed.
Climate redefined	Definition	Geographic physical exposure — coastal flood and storm-surge risk, wind-driven rain, and salt and wind weathering of building fabric — was folded into the existing node rather than added as a separate construct.
Climate → *IEQ* rationale extended	Rationale	Areas subject to driving rain are more prone to moisture ingress and therefore to damp dwellings, a mechanism distinct from the heat-demand route.
Construct definitions published	Instrument	Definitions were requested on the grounds that what a variable denotes determines the direction of its edges; a definitions view was added to the visualisation before the sign-off round.
Orientation-sensitivity analysis	Analysis	Several edges were argued to be plausibly reciprocal, with the requirement that the paper be explicit about the assumption each orientation encodes. The joint analysis reported in Section 4.2 was designed in response.

constructs have a defensible LSOA-level measure. Home energy efficiency is the median Standard Assessment Procedure (SAP) score of dwellings in the area (**?**), mental health is depression prevalence from Quality and Outcomes Framework registers (**?**) with the SAMHI (**?**) used in a robustness run, and *Fuel_Poverty* is the sub-regional Low Income Low Energy Efficiency proportion. The nine unmapped constructs are either latent by design (*Neighbourhood*, *U_AreaHistory*) or have no LSOA-level measure (*IEQ*, *Noise*, *Social_Capital*, *Digital_Capital*, *Policy_Scheme_Access*, *Isolation*, *Substance_Abuse*). Table 4 reports the mapping in full.

Each measured construct is represented here by a single variable, which is a narrower operationalisation than the one the companion paper uses for estimation, where the five adjustment-set constructs are represented jointly by 17 variables. The difference is required by the algorithms rather than chosen for convenience. PC and FCI treat every column as a node in the graph they search over, so entering several columns for one construct would test a different graph from the one under review: the construct would appear as a cluster of mutually dependent nodes, and every edge incident to it would be split across them.

Keeping one column per construct preserves the correspondence between node and construct, at the cost of discarding information and introducing single-proxy measurement error. We select the variable judged most representative of each construct, and address the cost directly in the composite sensitivity run, which replaces each variable with the first principal component of that construct's full variable block — recovering the multi-column operationalisation while keeping one node per construct. Because nine constructs are unmeasured, all checks are run on the subgraph induced over the 16 measured nodes, which retains 40 of the 89 edges.

3.5.2. Testing the graph's implied independencies

The induced subgraph implies a set of conditional independencies, one minimal d-separator per non-adjacent pair, following the basis `dagitty` uses for its `localTests` routine (**?**). Each is tested by partial correlation with a Fisher-z transformation. At $n \approx 32{,}000$ almost any departure from exact independence is statistically significant, so implications are judged by effect size — the absolute partial correlation — rather than by p-value, and we report the number failing at thresholds of 0.05, 0.10, and 0.15.

A failed implication does not by itself mean the graph is wrong, because the implication is a prediction of the subgraph rather than of the DAG. Removing the nine unmeasured constructs deletes every path that runs through them, so two variables the DAG connects only through an unmeasured node have no path left in the subgraph, which therefore predicts that they are independent. The DAG predicts the opposite. When the data show them to be dependent, the subgraph's prediction fails and the DAG's succeeds.

Separating the two cases is the point of the test. We annotate every failed implication with whether the pair is d-connected in the full 25-node DAG. Where it is, the failure is what the DAG predicts and the amputation obscured. Where it is not, the DAG separates the pair with every node present and the data still show dependence, which is evidence against the graph. Only the second category counts against it.

Figure 2a works one implication through in full, and the one chosen is not incidental: the graph's claim that education reaches HEE only through economic capital, housing and tenure is what makes $\mathcal{A}$ minimal, and it is directly testable. At $n \approx 32{,}000$ almost any exact null is rejected, so the verdict rests on the magnitude of the partial correlation rather than on its p-value.

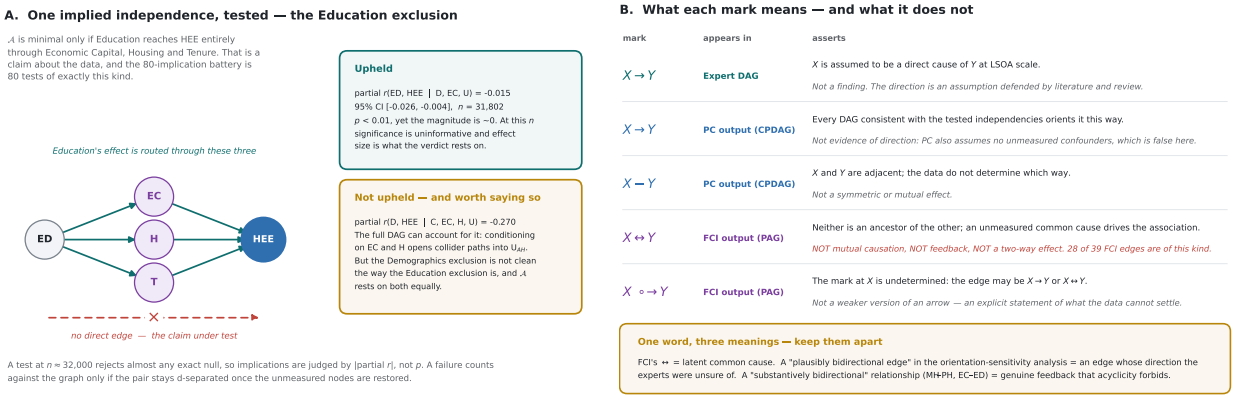


Figure 2: (a) One implied independence worked through: the claim that education reaches HEE only through economic capital, housing and tenure, and the partial correlation that tests it, shown against a second exclusion that behaves differently. (b) Every edge mark used in this paper, what it asserts, and what it does not. The bidirected arrow of a partial ancestral graph denotes an unmeasured common cause and not a reciprocal effect.

3.5.3. Structure recovery

Both algorithms proceed by deletion rather than construction, which is what makes the comparison with an expert graph meaningful. PC-stable begins from the complete graph over the 16 measured constructs and, for each pair, searches for a conditioning set that renders them independent; where one is found the edge is deleted, and the surviving adjacencies form the skeleton. Orientations are then imposed only where the

pattern of surviving independencies forces them. The Education–HEE pair is an instance drawn from this graph: a conditioning set exists, so no edge survives, which is also what the expert graph asserts (Figure 2a).

We ran PC-stable (**?**) at $\alpha \in \{0.01, 0.005, 0.001\}$, each configuration with and without background knowledge. The background knowledge encodes the expert causal ordering as tiers — *Climate* and *Urbanicity* exogenous through to MH last — which forbids edges running backwards through the ordering but leaves the skeleton untouched. Comparing the two isolates the contribution of the expert ordering to orientation.

PC assumes causal sufficiency, which nine unmeasured constructs violate by construction. We therefore also ran FCI (**?**) at the same three thresholds. FCI performs the same deletion step but does not assume that every relevant construct is measured. Where an association survives conditioning and no orientation consistent with a direct effect can be found, FCI records a bidirected edge ($\leftrightarrow$) in a partial ancestral graph, stating that the association is attributable to an unmeasured common cause rather than to an effect of either construct on the other. The proportion of bidirected edges is thus a direct measure of how far area-level associations in these data are driven by latent structure, and with nine constructs unmeasured by design FCI is the algorithm matched to the data.

Three distinct notions of a two-headed relationship arise in this paper and are kept apart throughout. A *bidirected edge* is FCI's $\leftrightarrow$ and denotes an unmeasured common cause. An *orientation-uncertain edge* is one whose direction the reviewing panel judged arguable, and which is therefore varied in the sensitivity analysis of Section 4.2. A *reciprocal relationship* is genuine mutual causation, which the acyclicity requirement forbids and which is discussed as a framework limitation in Section 5.2. None of the three implies either of the others. Figure 2b gives the full key to the edge marks used here, including what each does not assert.

Discovered graphs are compared to the expert induced subgraph on skeleton precision and recall for undirected adjacency, structural Hamming distance (SHD, counting missing, extra, and orientation-mismatched edges), and v-structure recall. We report two sensitivity analyses: substituting the SAMHI for depression prevalence, and replacing each single-column proxy with the first principal component of a multi-column block for that construct.

3.5.4. What would have counted as a contradiction

Because a corroborating result is easy to report and hard to interpret, we state the falsification conditions before the results. Two would have required the graph to be redrawn rather than defended.

First, an HEE–MH adjacency. The DAG asserts that the effect of energy efficiency on mental health is fully mediated, with no direct edge. Had discovery recovered an adjacency between HEE and MH that survived conditioning on the adjustment set, the graph's central structural claim would have been contradicted, and the mediated architecture — on which the choice of estimand in the companion paper depends — would require revision.

Second, disagreement concentrated among pairs that are d-separated in the full DAG. Discovery on a subgraph is expected to recover dependencies that latent projection induces; those failures are predicted by the graph. What the graph does not predict is empirical dependence between pairs it separates even with every node present. Had the failed implications been dominated by that category rather than by the latent-cut category, the natural reading would have been misspecification of the graph itself.

We also note in advance what no result of this kind could establish. Recovering an edge does not establish its direction, since orientations in a CPDAG or PAG are equivalence-class statements. Nor can these checks estimate the HEE–MH effect: the mediators that carry it are, with one exception, unmeasured. Analyses were run in Python using `causal-learn` for PC and FCI and `networkx` for d-separation, the canonical R toolchain being unavailable in the analysis environment; the algorithms are standard and the structures are expected to reproduce under `pcalg` and `dagitty`.

4. Results

4.1. Literature Support for the Causal DAG

Figure 3 presents the reviewed graph. The subsections below justify it construct by construct; the evidence for each individual edge, together with the citations supporting it, is given in Appendix A.

This section sets out the theoretical logic of the DAG linking home energy efficiency (HEE) to mental health (MH) at the LSOA level in England and Wales. The graph comprises 23 observed nodes and two

Table 4
LSOA-level variables used for the causal-discovery checks. One variable represents each measured construct, so that each node in the tested graph corresponds to one construct; the companion estimation paper instead represents the five adjustment-set constructs jointly by 17 variables. Proxy sign does not affect correlation-based independence tests or the recovered skeleton.

Construct	Variable	Definition
Measured (16)		
HEE	`RATING_MEDIAN_100`	Median EPC/SAP rating, 1–100
MH	`Dep rate`	QOF GP-recorded depression prevalence
Fuel_Poverty	`fuel_poor_pct`	LILEE: households in fuel poverty (%)
Economic_Capital	`Income Score (rate)`	IMD income-deprivation rate
Education	`Education, Skills and Training Score`	IMD education-deprivation score
Housing_Char.	`BUILT_BEFORE_1950`	Share of dwellings built pre-1950
Tenure	`rented`	Rented share, private and social
Demographics	`65+`	Share of residents aged 65 and over
Urbanicity	`pop_density`	Population density, persons per hectare
Climate	`DJF_w`	Mean winter temperature
Air_Pollution	`no2_mean`	Mean NO_2 concentration
Green_Space	`green_act`	Active green-space access
Overcrowding	`hhsize_ave`	Average household size
Neighbourhood_Stress	`Crime Score`	IMD crime-deprivation score
Accessibility	`gpp_dist`	Distance to nearest GP practice
PH	`Health Deprivation and Disability Score`	IMD health-deprivation and disability score
Unmeasured (9)		
Neighbourhood	—	Latent scale-bridging construct by design
U_AreaHistory	—	Latent root cause by design
IEQ	—	No area-level damp, mould, or thermal-comfort measure
Noise	—	No environmental-noise (L_{den}/L_{night}) measure
Social_Capital	—	No civic-participation measure
Digital_Capital	—	No broadband or digital-exclusion measure
Policy_Scheme_Access	—	No scheme-uptake measure
Isolation	—	No isolation or loneliness measure
Substance_Abuse	—	No misuse-prevalence measure

latent constructs (*U_AreaHistory* and *Neighbourhood*) connected by 89 directed edges, and is shown in panel (a) of Figure 5. The review is organised into seven thematic groupings: upstream socioeconomic determinants; housing, tenure, and overcrowding; determinants of the exposure; indoor environmental quality and fuel poverty as proximal mediators; environmental mediators; social and psychological mediators; and structural and exogenous variables. Source-level evidence for each edge is given in the edge-by-edge tables of Appendix A; this subsection addresses the architectural role of each node group and the design choices that are not visible at edge level. A final subsection documents edges considered but excluded (Section 4.1.8).

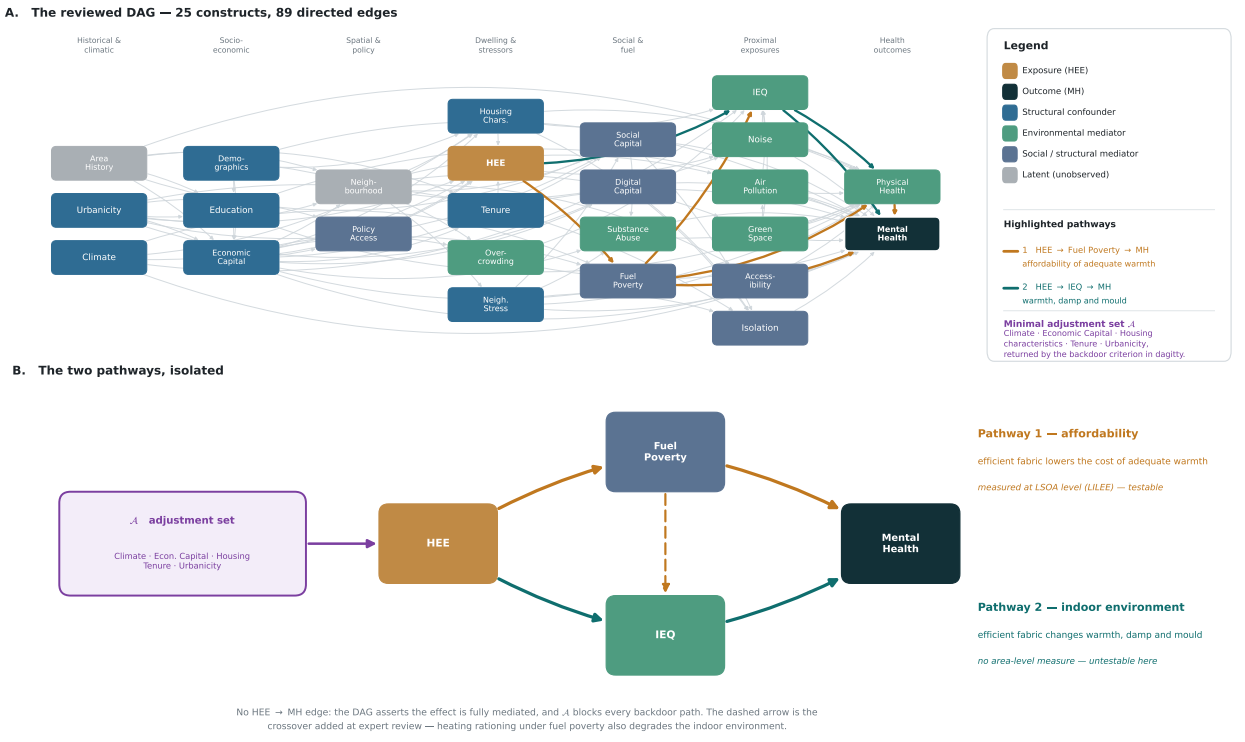


Figure 3: (a) The reviewed DAG: 25 constructs and 89 directed edges, arranged in the seven domain columns used for expert review and coloured by hypothesised causal role, with the two channels carrying the whole HEE→MH effect highlighted. (b) Those two channels in isolation, with the minimal adjustment set $\mathcal{A}$ blocking the backdoor. *Fuel_Poverty* has an LSOA-level measure and *IEQ* does not, which is why only one of the two channels can be tested in Section 4.3. The dashed arrow is the crossover added at expert review.

4.1.1. Upstream Socioeconomic Determinants

Economic_Capital, *Education*, and *Demographics* occupy the most upstream observed positions, motivated by fundamental-cause theory: socioeconomic position embodies flexible resources that protect health through multiple substitutable mechanisms (**??**). All three influence both exposure and outcome, but the DAG routes their effects on HEE through dwelling and economic mediators rather than as direct edges (Section 4.1.8). Sibling fixed-effects evidence shows that the OLS association between education and depressive symptoms attenuates by 40–70% once unobserved family-level confounders are controlled, consistent with education operating partly through economic and partly through non-economic channels (**?**). A direct consequence of this mediation structure is that *Demographics* and *Education* do not enter the minimal adjustment set $\mathcal{A}$, because the backdoor paths they would otherwise close are blocked at the mediators through which their effects are routed (Section 4.2).

4.1.2. Housing, Tenure, and Overcrowding

The housing cluster is modelled through three conceptually distinct nodes. *Housing_Characteristics* is defined strictly as physical dwelling form — age, construction era, type, and materials — and is the dominant determinant of the baseline EPC attainable through retrofit (**??**). The node is deliberately narrower than housing quality: damp, mould, cold, and indoor pollutants belong to *IEQ*, so the prospective evidence that persistent poor housing predicts distress (**??**) is routed through *Housing_Characteristics* → *IEQ* → *MH* rather than as a direct edge. *Tenure* enters through the landlord–tenant split incentive that suppresses retrofit investment in the private rented sector (**??**) and through tenure insecurity as an independent stressor (**?**). *Overcrowding* reaches MH directly, through loss of privacy and interpersonal friction (**?**), and indirectly through IEQ degradation via elevated moisture loads and reduced per-occupant ventilation.

4.1.3. Determinants of the Exposure: Home Energy Efficiency

HEE has five parents in the DAG: *Economic_Capital*, *Housing_Characteristics*, *Tenure*, *Policy_Scheme_Access*, *Urbanicity*, and *Climate*. The first three operate through affordability, technical feasibility, and split incentives. *Policy_Scheme_Access* represents area-level uptake of named government schemes — principally the Energy Company Obligation, which accounts for the majority of measures installed under UK government programmes (**?**), alongside the Warm Home Discount (**?**) and the now-closed Green Deal (**?**). Defining the node around scheme uptake rather than "policy" in the abstract clarifies what is modelled causally: households are exposed to specific schemes with defined eligibility rules. *Urbanicity* and *Climate* shape HEE through the spatial distribution of dwelling stock and the regional structure of heat demand (**??**). The parents of HEE anchor the minimal adjustment set: $\mathcal{A}$ = {Climate, Economic_Capital, Housing_Characteristics, Tenure, Urbanicity}.

4.1.4. Indoor Environmental Quality as the Proximal Mediator

IEQ is the principal proximal mediator on the pathway from HEE to MH, and the mechanism is non-monotonic: retrofits including adequate ventilation produce net indoor-environment gains, while envelope improvements without ventilation provision can trap moisture and pollutants and worsen conditions relative to baseline (**????**). Modelling IEQ as a mediator rather than assuming a uniformly beneficial HEE effect is what captures this tension. The IEQ–MH edge rests on converging evidence that damp, mould, cold, and indoor pollutants predict depression, anxiety, and perceived loss of control (**???**), complemented by the physical-health channel $IEQ \rightarrow PH \rightarrow MH$. IEQ also absorbs several exogenous influences — air-pollution infiltration (**?**), climate-driven indoor overheating (**?**), noise ingress (**?**), and overcrowding-driven moisture loads — that would otherwise require direct environmental-to-MH edges with weaker mechanistic support.

Fuel_Poverty, added following expert review, is the second proximal mediator and carries the affordability channel. The node is operationalised through the Low Income Low Energy Efficiency (LILEE) indicator and is jointly determined by dwelling efficiency ($HEE \rightarrow FP$), household economic resources ($Economic_Capital \rightarrow FP$), and the climate-driven structure of heat demand ($Climate \rightarrow FP$). Its downstream edges reach mental health directly, through the persistent worry, financial strain, and heat-or-eat trade-offs documented in the fuel-poverty literature (**???**), and indirectly through physical health and indoor conditions: households rationing heating under-heat rooms and ventilate less, promoting condensation, damp, and mould independently of the dwelling fabric ($FP \rightarrow IEQ$; **?**). Distinguishing this behavioural-affordability route from the physical-fabric route matters for identification: both are descendants of HEE and must not be conditioned upon when estimating the total effect, but they imply different policy instruments — fabric retrofit versus bill support — whose relative contributions a mediation analysis could in principle separate.

4.1.5. Environmental Mediators

Air_Pollution, *Green_Space*, and *Noise* are area-level environmental mediators whose distribution is set at neighbourhood scale and whose effects reach MH directly and through IEQ and PH. Evidence strength varies: the air-pollution–depression association is supported by multiple meta-analyses (**???**); green-space effects are supported by large UK longitudinal evidence (**??**); the noise–MH association is less settled, with non-significant confidence intervals in the road-traffic meta-analysis and stronger effects isolated to aircraft noise (**??**). *Noise* also creates a secondary HEE–MH pathway: double and triple glazing and improved airtightness attenuate external noise transmission (**??**), making noise reduction an identifiable retrofit co-benefit alongside thermal comfort.

4.1.6. Social and Psychological Mediators

Five nodes capture social and psychological mechanisms: *Social_Capital*, *Isolation*, *Digital_Capital*, *Neighbourhood_Stress*, and *Substance_Abuse*. *Social_Capital* is an intrinsically area-level construct representing civic participation and associational density (**?**), distinct from individual-level *Isolation*, which captures geographic and relational disconnection (**??**). *Neighbourhood_Stress* bundles chronic crime exposure, disorder, and perceived unsafety into a single stressor anchored empirically in the crime literature (**?**); its distinct downstream effects on PH (through allostatic load), *Isolation* (through withdrawal), *Social_Capital* (through eroded collective efficacy), and *Substance_Abuse* (through self-medication and drug-market density) are documented in Table 13. *Digital_Capital* carries a single directed edge to MH

on the grounds that at LSOA aggregate level the access-to-services channel dominates the known adverse effects of digital overuse (??); this is a defensible but contestable choice, and the underlying relationship is reciprocal. *Substance_Abuse* enters as both a consequence of upstream deprivation and a cause of further deterioration via PH and MH (?).

4.1.7. Structural and Exogenous Variables

Three nodes occupy structurally distinct positions. *Urbanicity* is a macro-level categorical variable cascading through economic opportunity, educational access, social capital, and digital infrastructure. It is not a synonym for deprivation: many rural areas are affluent and many urban areas deprived, and the urban–rural dimension sits upstream of deprivation as a determinant of housing stock, amenity distribution, and infrastructure provision.

Neighbourhood is a latent scale-bridging construct rather than a deprivation proxy or a spatial container. The node marks the transition from household-level to area-level causation: the edges leaving it (Table 12) are area-level exposures — *Air_Pollution*, *Green_Space*, *Noise*, *Accessibility*, *Neighbourhood_Stress*, *Housing_Characteristics*, and *Social_Capital* — that aggregate upstream household-level and historical factors into place-based exposures. Quasi-experimental evaluations of UK traffic and low-emission interventions (???) support a causal reading of this aggregation channel for air pollution specifically; for other area-level exposures the mechanism rests on weaker, primarily cross-sectional evidence, an asymmetry acknowledged in Section 5.2. *Neighbourhood* is latent because it has no satisfactory direct operationalisation at LSOA level; its role is to represent a multi-level structure parsimoniously without positing a specific area-level variable.

Climate is exogenous, influencing HEE demand profiles (?), housing stock integrity (?), IEQ under extreme heat (?), and MH directly through extreme-weather trauma (?). *U_AreaHistory* captures the unmeasured legacy of historical planning, deindustrialisation, and structural disinvestment. Its inclusion as a latent root cause is justified by the stability of area-level deprivation over half a century ($r = 0.706$ between 1971 and 2021 deprivation rankings in England and Wales; ?). It has no incoming edges and lies on no open backdoor path for the HEE–MH estimand; identification does not require conditioning on it, but does require its effects on HEE and MH to be transmitted entirely through the observables in the adjustment set (Section 5.2).

4.1.8. Edges Considered But Excluded

The DAG represents the causal structure we judged most defensible on current evidence, but several edges a reader might expect were considered and deliberately omitted. Transparency about exclusions is essential for reproducible DAG-based inference: ? observe that most published DAGs report only included edges, leaving readers unable to distinguish edges evaluated and rejected from edges never considered. Table 16 documents each excluded edge, the prima facie case for inclusion, and the reason for omission. Three mechanisms account for the seven exclusions.

The first, most consequential for identification, is that the effect an edge would represent is already transmitted through an explicit mediating pathway, so a direct edge would double-count it and bias the estimation of mediator contributions. Four exclusions fall here: *Education* → *HEE* and *Demographics* → *HEE*, whose effects operate through *Economic_Capital*, *Housing_Characteristics*, and *Tenure* once income and dwelling characteristics are conditioned on (??); *Housing_Characteristics* → *MH*, whose prospective evidence base (??) reflects housing quality effects now routed through *Housing_Characteristics* → *IEQ* → *MH*; and *HEE* → *Economic_Capital*, whose bill-saving channel is represented within the HEE–MH pathway rather than as a modifier of the confounder set, avoiding conflation of the exposure with one of its consequences.

The second mechanism is acyclicity. Two exclusions correspond to reciprocal relationships for which DAG formalism requires a single direction: *Economic_Capital* → *Education*, where the adult labour-market returns channel (*Education* → *Economic_Capital*) is retained as the dominant cross-sectional data-generating process at working age, with intergenerational persistence partially absorbed by *U_AreaHistory*; and *MH* → *PH*, where the reverse direction is retained because *PH* → *MH* is the direction that threatens identification through backdoor paths. In both cases the omitted direction reflects a real mechanism, and its exclusion is a framework limitation noted in Section 5.2 rather than a denial of the relationship.

The third mechanism is node definition: the excluded *Neighbourhood_Stress* → *Green_Space* edge would require redefining *Green_Space* to include behavioural engagement rather than area-level provision, breaking consistency with its operationalisation elsewhere. The mental health effect it would carry is already captured through *Neighbourhood_Stress* → *Isolation* → *MH* and the direct *Neighbourhood_Stress* → *MH* channel.

A notable consequence of the first mechanism is that *Demographics* and *Education* do not enter the minimal adjustment set, because the backdoor paths they would close are blocked at the mediators through which their effects are routed. The implications for the adjustment set are addressed in Section 4.2. In all cases, omission is a modelling choice with identified substantive implications, not an oversight.

4.2. Minimal Adjustment Set

Given the structure articulated in Section 4.1, identifying the total causal effect of HEE on MH requires conditioning on observed variables sufficient to block all backdoor paths between exposure and outcome. Applying the backdoor criterion (**?**) via the `dagitty` R package (**?**) returns a single minimal sufficient adjustment set:

$$\mathcal{A} = \{\text{Climate, Economic_Capital, Housing_Characteristics, Tenure, Urbanicity}\}. \tag{1}$$

All other observed nodes fall into one of three disqualifying categories: mediators on the directed path from HEE to MH (*IEQ*, *Fuel_Poverty*, *Noise*, *PH*); descendants of HEE that are not mediators and would block no additional backdoor path; and non-descendants whose backdoor paths to HEE are already blocked by conditioning on $\mathcal{A}$. The set was re-derived after every structural revision arising from expert review, including the addition of the *Fuel_Poverty* mediator, and is unchanged by them. Expert review identified eight edges whose orientation it judged arguable, and a joint sensitivity analysis over all $3^8 = 6561$ configurations of those edges — each held at its committed orientation, reversed, or replaced by a latent common cause — is reported alongside the edge tables in Appendix A. Of these, 73.7% are cyclic and therefore inadmissible, 18.9% are identifiable, and 7.4% are non-identifiable. Equation 1 is returned under the committed orientation, and a single assumption accounts for every failure: a latent common cause of HEE and *Economic_Capital* is present in all 486 non-identifiable configurations. Identification therefore rests on the absence of that specific latent structure rather than on the eight orientations jointly, which is a narrower and more checkable claim. It is also the same relationship the discovery checks could not recover (Section 4.3), so two independent lines of evidence locate the weakest point of the design at the same edge. The latent nodes *U_AreaHistory* and *Neighbourhood* are by construction unavailable for conditioning; identification requires instead that their effects on HEE and MH are transmitted entirely through the observables in $\mathcal{A}$, an assumption discussed in Section 5.2.

A reader familiar with regression-based confounder selection will observe that $\mathcal{A}$ coincides exactly with the parents of HEE — the variables whose arrows point directly into the exposure. If the adjustment set is simply the parents of the exposure, what work is the DAG doing that listing HEE's immediate predictors could not have done? The coincidence is real; the inference that the DAG is redundant is not, for three reasons.

First, the parents-of-exposure criterion is a theorem of DAG-based identification, not a pre-DAG intuition. The parent-adjustment result (**?**) guarantees that conditioning on the parents of the exposure identifies the total effect *provided the graph is correctly specified and acyclic, and no parent is itself a descendant of the exposure*. Establishing that these conditions hold requires exactly the edge-by-edge reasoning of Section 4.1 and Appendix A. Without the DAG, identifying $\mathcal{A}$ by inspection assumes what the DAG is constructed to prove.

Second, $\mathcal{A}$ is minimal only because several candidate confounders were excluded from the direct-edge set on principled grounds (Section 4.1.8). *Education* and *Demographics* are the consequential cases: both would have entered a pre-DAG regression as obvious confounders, and both would appear in an adjustment set derived from a DAG that included direct edges from them to HEE. Their absence is a finding of the DAG-construction process, not an input to it, resting on the evidence in Appendix A and Table 16 that their effects on HEE are routed through *Economic_Capital*, *Housing_Characteristics*, and *Tenure*.

Third, the DAG specifies what *not* to condition on, which a parents-of-exposure heuristic cannot. *IEQ* is the proximal mediator and a natural candidate for inclusion in an atheoretical regression; conditioning on it would block the causal channel the analysis is designed to estimate and bias the total-effect estimand

toward zero. *Overcrowding*, *PH*, and the environmental mediators are excluded on the same grounds. The distinction between confounders that must be adjusted for and mediators that must not is not recoverable from variable labels alone.

The parsimony of $\mathcal{A}$ is therefore a feature of the analysis rather than a symptom of redundancy. The DAG produces a defensible, minimal, and non-obvious identification strategy whose transparency is itself the principal methodological contribution.

4.3. Causal-Discovery Checks

4.3.1. Graph and sample

The canonical graph was verified before testing: 25 nodes, 89 directed edges, acyclic. Sixteen constructs have an LSOA-level proxy (Table 4); the subgraph induced over them retains 40 of the 89 edges. All checks below use $n = 31{,}802$ areas.

4.3.2. Implied conditional independencies

The induced subgraph implies 80 testable conditional independencies. Judged by effect size, 69 fail at $|r| \geq 0.05$, 62 at $|r| \geq 0.10$, and 55 at $|r| \geq 0.15$; the median absolute partial correlation across the 80 is 0.21 and the maximum is 0.64. Taken at face value this is a poor fit. The count on its own is uninformative, however, because most of these implications are predictions of the subgraph that the full DAG never made.

Of the 62 failures at the 0.10 threshold, 48 (77%) involve pairs that are d-connected in the full 25-node DAG. The subgraph implied independence only because removing the unmeasured nodes cut the path between them. The strongest failure is representative: the subgraph implies that air pollution and urbanicity are independent, while the full graph routes *Urbanicity* $\rightarrow$ *Neighbourhood* $\rightarrow$ *Air_Pollution*, and the observed $r = 0.64$ is that latent path resurfacing. These failures are predicted by the graph rather than evidence against it.

The remaining 14 are genuine tensions — d-separated in the full DAG yet empirically dependent. Eight involve *Climate* (with *Urbanicity*, *Air_Pollution*, *Tenure*, *Neighbourhood_Stress*, *Accessibility*, *Overcrowding*, *Demographics*, and *Green_Space*). Of the remainder, the largest is *Demographics*–*Urbanicity* ($r = 0.53$), consistent with rural areas skewing older through a shared settlement-history cause of the kind *U_AreaHistory* already represents; the others involve the household-size proxy for overcrowding. We interpret these as proxy and aggregation artefacts rather than missing edges, for reasons set out in Section 5.1.

The battery also contains direct tests of the two exclusions on which the minimality of $\mathcal{A}$ rests (Section 4.2), and they do not behave alike. The implication that education is independent of HEE given demographics, economic capital and urbanicity holds: the partial correlation is -0.015 (95% CI -0.026 to -0.004), negligible at this sample size. The corresponding implication for demographics, given climate, economic capital, housing characteristics and urbanicity, does not: the partial correlation is -0.270. The full graph can account for the second, because conditioning on *Economic_Capital* and *Housing_Characteristics* opens collider paths into *U_AreaHistory*, and the pair is d-connected once the latent nodes are restored. The magnitudes nonetheless differ by more than an order of magnitude, so the exclusion of *Demographics* carries more of the identification burden than the exclusion of *Education*, and is correspondingly more exposed to misspecification of the latent structure.

4.3.3. Structure recovery

Table 18 reports agreement between the discovered graphs and the expert induced subgraph. PC-stable recovers 43–45 edges depending on threshold, with skeleton precision 0.41–0.42 and recall 0.45–0.47. Adding the expert causal ordering as background knowledge leaves the skeleton unchanged, as expected, but changes the orientations substantially: of the 19 adjacencies PC and the expert graph share, PC reverses 14 and leaves one undirected when run on the data alone, against two reversed once the ordering is supplied. Structural Hamming distance falls correspondingly from 62 to 50 and v-structure recall rises from 0.01 to 0.14. The expert ordering therefore does the orientation work that the data alone do not.

FCI recovers 37–39 edges with comparable skeleton agreement. At $\alpha = 0.01$, 28 of its 39 edges are bidirected — FCI's statement that an association is attributable to an unmeasured common cause rather than a direct effect. Every edge incident to HEE is bidirected (*Accessibility*, *Fuel_Poverty*, *Green_Space*, *Housing_Characteristics*, and *Tenure*). Figure 5 shows the three structures on a shared layout.

Every edge at HEE, and what each graph says about it

PC-stable and FCI at $\alpha = 0.01$, depression prevalence, $n = 31{,}802$ LSOAs. Rows are the union of HEE-incident edges across the three graphs.

neighbour of HEE	Expert DAG	PC-stable		FCI
Accessibility	*not recovered*	HEE → A		**A ↔ HEE**
Climate (in $\mathcal{A}$)	C → HEE	*not recovered*		*not recovered*
Economic Capital (in $\mathcal{A}$)	EC → HEE	*not recovered*		*not recovered*
Fuel Poverty	HEE → FP	HEE → FP		**FP ↔ HEE**
Green Space	*not recovered*	HEE → GS		**GS ↔ HEE**
Housing Characteristics (in $\mathcal{A}$)	H → HEE	HEE → H	*direction reversed*	**H ↔ HEE**
Tenure (in $\mathcal{A}$)	T → HEE	HEE → T	*direction reversed*	**T ↔ HEE**
Urbanicity (in $\mathcal{A}$)	U → HEE	*not recovered*		*not recovered*
Mental Health	*not recovered*	*not recovered*		*not recovered*

What the row pattern shows

1. HEE and MH are non-adjacent in all three graphs — the pre-stated falsification condition, not triggered.
2. Three of the five adjustment-set edges into HEE (Climate, Economic Capital, Urbanicity) are recovered by neither algorithm: the check is silent on exactly the edges identification depends on.
3. PC reverses two of the expert edges it does recover, and FCI marks all five HEE edges latent-confounded — which is why no orientation reported here is read as evidence about causal direction.

Figure 4: Every edge incident to HEE in any of the three graphs, and what each graph says about it, at $\alpha = 0.01$ with depression prevalence as the outcome. Rows are the union of HEE-incident adjacencies; the HEE–MH row is the falsification condition stated in Section 3.5.

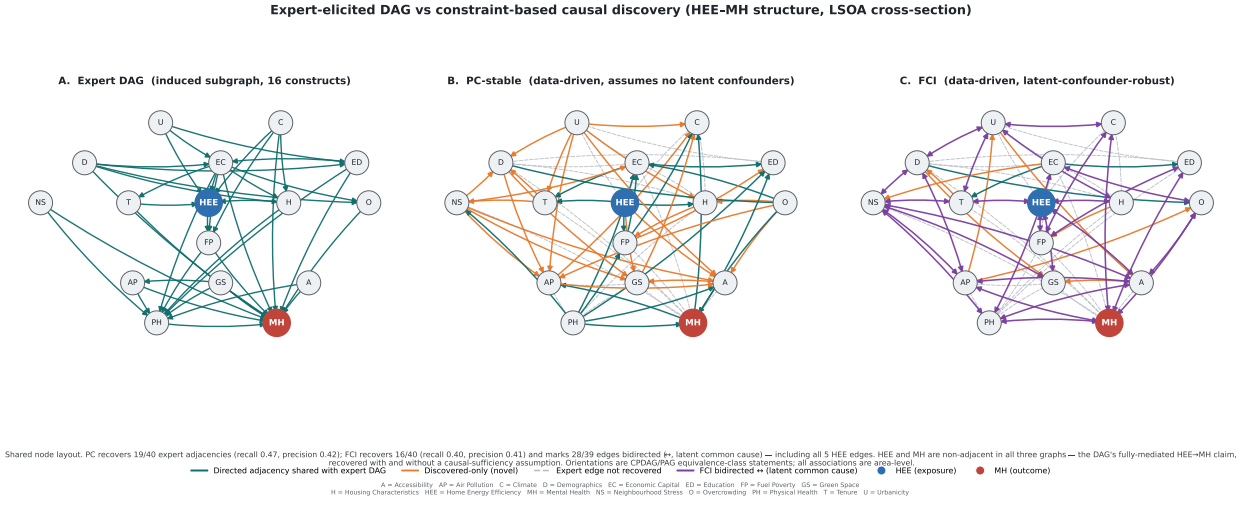


Figure 5: The expert DAG and the discovered structures on a shared node layout, over the 16 constructs with an LSOA-level proxy ($n = 31{,}802$; MH = depression prevalence). (a) The expert DAG's induced subgraph, 40 directed edges. (b) PC-stable at $\alpha = 0.01$, which assumes no latent confounders: it recovers 19 of the 40 expert adjacencies (recall 0.47, precision 0.42). (c) FCI at $\alpha = 0.01$, robust to latent confounding: it recovers 16 of 40 (recall 0.40, precision 0.41) and marks 28 of its 39 edges bidirected (↔, denoting an unmeasured common cause), including all five HEE edges. HEE and MH are non-adjacent in all three graphs. Orientations in (b) and (c) are equivalence-class statements, and all associations are area-level.

Table 5
Partial correlations for the fuel-poverty mediation chain, $n = 31{,}802$. $\mathcal{A}$ denotes the minimal adjustment set of Equation 1. Mental health is coded so that higher values are worse; a negative HEE–FP correlation therefore means more efficient stock is associated with less fuel poverty.

Relationship	Partial r	Recovered by PC
HEE $\sim$ FP (marginal)	-0.33	—
HEE $\sim$ FP $\mid \mathcal{A}$	-0.39	Yes, oriented HEE $\rightarrow$ FP
FP $\sim$ MH (marginal)	$+0.073$	—
FP $\sim$ MH $\mid \mathcal{A}$	$+0.010$	No
HEE $\sim$ MH $\mid \mathcal{A}$	$+0.062$	No (non-adjacent)
HEE $\sim$ MH $\mid \mathcal{A}$, FP	$+0.071$	No (non-adjacent)

Across every configuration — three thresholds, with and without background knowledge, PC and FCI alike — HEE and MH are non-adjacent. The edge-level pattern is more informative than the aggregate counts. In the expert subgraph, HEE's neighbourhood is {*Climate*, *Economic_ Capital*, *Fuel_ Poverty*, *Housing_ Characteristics*, *Tenure*, *Urbanicity*}. PC recovers the adjacencies to *Fuel_ Poverty*, *Housing_ Characteristics* and *Tenure*, adds *Accessibility* and *Green_ Space*, and misses *Climate*, *Economic_ Capital* and *Urbanicity*. Two of the three recovered adjacencies are oriented against the expert graph, as HEE $\rightarrow$ *Housing_ Characteristics* and HEE $\rightarrow$ *Tenure*; FCI marks all five of its HEE adjacencies bidirected. Neither pattern is treated as evidence about direction, for the reason given in Section 3.5, and both are shown row by row in Figure 4. The three missed adjacencies are the more consequential result: *Climate*, *Economic_ Capital* and *Urbanicity* are three of the five members of $\mathcal{A}$, so discovery is silent on precisely the edges identification depends on, and the corroboration it offers does not extend to them. Most edges PC adds sit among the environmental and social mediators that share the single latent parent *Neighbourhood* (*Accessibility–Air_ Pollution*, *Accessibility–Green_ Space*, *Air_ Pollution–Neighbourhood_ Stress*, and others). Marginalising a common parent out of a graph induces exactly these dependencies. Table 19 lists the full edge-level disagreement.

4.3.4. Fuel poverty: the one testable mediator

Fuel_ Poverty is the only mediator on the HEE–MH pathway with an area-level measure, which makes the chain from energy efficiency through fuel poverty to mental health testable for the first time. The two halves behave differently (Table 5). The upstream half is recovered strongly: the HEE–FP partial correlation is -0.33 marginally and strengthens to -0.39 conditional on $\mathcal{A}$, and PC orients the edge *HEE* $\rightarrow$ *FP*. The downstream half is not recovered: the FP–MH correlation of $+0.073$ falls to $+0.010$ (95% CI -0.001 to 0.021) once $\mathcal{A}$ is conditioned on, and the edge is absent from every discovered graph. Conditioning on *Fuel_ Poverty* does not reduce the HEE–MH partial correlation, which moves from $+0.062$ to $+0.071$.

At the ecological, cross-sectional level, then, fuel poverty carries no residual association with mental health once area deprivation is accounted for, and it is not the channel transmitting the small HEE–MH residual. FCI marks fuel poverty's own edges as latent-confounded (*Climate* $\leftrightarrow$ *FP*, *Education* $\leftrightarrow$ *FP*, *FP* $\leftrightarrow$ *HEE*).

4.3.5. Sensitivity

Substituting the SAMHI for depression prevalence reproduces the pattern: 80 implications with 62 failures at $|r| \geq 0.10$, 77% of them latent-cut, PC skeleton precision and recall of approximately 0.38 and 0.40, 24 bidirected FCI edges, HEE and MH non-adjacent, and *FP* $\rightarrow$ *MH* again unrecovered. Replacing each single variable with the first principal component of that construct's full variable block — the multi-column operationalisation used for estimation, collapsed so that one node still corresponds to one construct — gives 67 of 80 implications failing at $|r| \geq 0.10$, 78% latent-cut, approximately 19 bidirected FCI edges, and HEE and MH again non-adjacent; *Climate* continues to dominate the genuine tensions, joined by proxy links between fuel poverty and tenure, housing, and overcrowding. We report the single-column run as primary for comparability and the composite as sensitivity.

4.3.6. Verdict against the pre-stated conditions

Neither falsification condition of Section 3.5 was triggered. HEE and MH are non-adjacent in every discovered graph at every threshold and under both outcome definitions, so the fully mediated architecture stands. Failed implications are dominated by the latent-cut category, 48 of 62, rather than by pairs the full graph separates. The 14 genuine tensions are concentrated on a single construct rather than distributed across the graph, which points to that construct's proxy rather than to the structure.

5. Discussions

5.1. Interpreting the Discovery Checks

Three findings bear on the graph's credibility, and each needs to be stated at the strength the evidence supports.

The first is the HEE–MH non-adjacency. The DAG asserts that energy efficiency reaches mental health only through mediators, and no discovered graph contradicts it, at any threshold, under either outcome definition, and whether or not causal sufficiency is assumed. The claim was falsifiable: a recovered HEE–MH edge surviving adjustment would have implied either a direct pathway the graph omits or a confounder outside $\mathcal{A}$, and would have required the graph to be redrawn. It held. The test is not decisive on its own, because most of the mediators carrying the effect are unmeasured, but it is the strongest structural check these data permit.

The second is the latent structure. FCI's finding that 28 of 39 edges are bidirected, including every edge incident to HEE, is the most consequential result for the companion estimation paper. It states that at this spatial scale, associations between area characteristics are dominated by unmeasured common causes rather than by direct effects. This is the premise on which the DAG's two latent nodes and its adjustment set rest, and the data supply it independently of the graph. It is also a warning against reading any area-level association as an effect — the caution that motivates the DAG in the first place.

The third is the failure pattern. Of the 62 implications the data reject, 48 are pairs the full DAG connects through an unmeasured node. Those dependencies are what the DAG predicts; it was the subgraph, stripped of the unmeasured nodes, that predicted independence. The graph is therefore not contradicted by the bulk of its apparent misfit, and the 14 remaining failures are the ones that test it. One qualification applies to reading this as support rather than as absence of contradiction: the annotation relies on the pathways the graph posits through *Neighbourhood* and *U_ AreaHistory* being correct.

Eight of the 14 remaining failures involve *Climate*, measured as mean winter temperature. That measure varies smoothly with latitude and region, so it correlates with most area characteristics through geography rather than through any edge the DAG draws. This is spatial confounding in the proxy, compounded by the coarseness of LSOA aggregation, and it is a limitation of the measurement rather than a defect in the construct: the claim that regional heat demand shapes both dwelling efficiency and the conditions of habitation does not require winter temperature to be conditionally independent of crime deprivation. Resolving these tensions requires a better climate measure and a model with an explicit spatial component. The same applies to *Overcrowding*, measured here by mean household size.

The fuel-poverty result is a substantive negative finding. The DAG routes part of the HEE–MH effect through fuel poverty, and the affordability channel has strong support in the individual-level literature (**???**). In these data the upstream half of that chain is strong and recovered, while the downstream half vanishes on adjustment for area deprivation. A cross-sectional ecological design is the wrong instrument for detecting mediation of this kind: fuel poverty at area level is largely a restatement of area deprivation, which the adjustment set already absorbs. This does not license removing the $FP \to MH$ edge, whose warrant is individual-level. It does indicate that the remaining unmeasured ecological mediators would behave the same way, which bounds what any area-level mediation analysis in this setting could recover.

The orientation result bears on the standing of expert knowledge in this design. Encoding the expert causal ordering as tiers left the skeleton untouched while reversing the direction of 12 of the 19 shared adjacencies, and raised v-structure recall from 0.01 to 0.14. The failure of data-only discovery here is not reticence but confidence: PC returns 44 directed edges and one undirected at $\alpha = 0.01$, orienting almost everything and agreeing with the expert graph on almost none of it. A method that commits to orientations it cannot support is a weaker basis for causal structure than one that declines to orient, and the ordering

that domain review supplied is what does the orienting. That is an argument for expert-elicited structure as a complement to discovery rather than an alternative to it.

5.2. Ecological Inference, Unmeasured Confounding, and the Limits of Area-Level Evidence

A further set of constraints concerns the data rather than the algorithms. The analysis is conducted at the area level, so every association reported here relates aggregate characteristics of neighbourhoods, not the energy efficiency of a dwelling to the mental health of its occupant. Associations observed at aggregate level cannot be straightforwardly attributed to the individual-level mechanism of interest (**?**). An area with a high median SAP score contains dwellings across the full efficiency distribution, and its residents vary correspondingly in their exposure to inefficiency, fuel poverty, and the thermal conditions of their homes. This limitation is inherent to the spatial scale of the available data and is shared by the small-area epidemiological literature on housing and health (**?**). It does not disqualify area-level analysis — much of the most policy-relevant evidence on health inequalities operates at this scale (**?**) — but it imposes a ceiling on causal claims and underscores the value of individual-level data linkage as a longer-term priority.

The strength of the causal evidence behind individual edges is also not uniform. Several edges, particularly those linking upstream factors to area-level exposures through the latent *Neighbourhood* node, are supported by quasi-experimental evaluations with credible identification: the London Ultra Low Emission Zone and Birmingham Clean Air Zone analyses use augmented synthetic control (**??**), and the London Low Traffic Neighbourhood evaluation uses difference-in-differences (**?**). Others, particularly the *Neighbourhood* $\rightarrow$ *Housing_Characteristics* and *Neighbourhood* $\rightarrow$ *Noise* pathways, rest on cross-sectional associations or mechanistic plausibility. The DAG treats all edges as causal for identification purposes, so this asymmetry is a qualitative limitation on the uniformity of causal claims across the graph, flagged in the edge-by-edge tables (Appendix A).

On unmeasured confounding, the DAG was not constructed casually. The structure was developed through systematic synthesis of 53 peer-reviewed publications spanning environmental epidemiology, housing studies, social epidemiology, and mental health research, and was then subjected to independent expert review. Every edge had to satisfy temporal precedence, mechanistic plausibility, and empirical support before inclusion (**?**), and the adjustment set was derived formally using the backdoor criterion. This compares favourably with the implicit, data-driven covariate selection characterising much of the prior housing–health literature (**?**). Four residual concerns nonetheless remain.

First, the proxies available to operationalise the nodes in $\mathcal{A}$ measure the underlying constructs imperfectly. Economic capital is operationalised through Income and Employment Deprivation scores from the Index of Multiple Deprivation, which capture relative area-level disadvantage but not the wealth, savings, and financial resilience that constitute economic capital as a causal construct. Where proxy measurement is imperfect, conditioning on the observed variable may not fully block the backdoor path through the latent construct, leaving residual confounding proportional to the measurement error (**?**). The discovery checks sharpen this concern rather than resolving it: the genuine tensions in Section 4.3 are concentrated in exactly the constructs whose proxies are weakest.

Second, identification relies on two mediation assumptions. *U_AreaHistory* is modelled as a root cause whose effects on HEE and MH are transmitted entirely through the observed nodes in $\mathcal{A}$ and the documented mediators. If historical area legacies influence present-day HEE or MH through channels not routed through those observed descendants, $\mathcal{A}$ will not block the relevant backdoor paths. An analogous assumption attaches to the exclusion of direct *Education* $\rightarrow$ *HEE* and *Demographics* $\rightarrow$ *HEE* edges (Section 4.1.8): $\mathcal{A}$ is minimal only if their effects on HEE are genuinely and completely routed through *Economic_Capital*, *Housing_Characteristics*, and *Tenure*. Residual direct channels — an energy-literacy pathway from education to retrofit uptake independent of income, or a discrimination-mediated pathway from demographics to scheme access not captured by *Policy_Scheme_Access* — would constitute omitted confounders. The exclusion rationale rests on evidence that these channels are small once the named mediators are conditioned on (**??**), but the assumption is not directly testable at this scale.

Third, the DAG framework imposes acyclicity on genuinely reciprocal relationships. The excluded *MH* $\rightarrow$ *PH* and *Economic_Capital* $\rightarrow$ *Education* edges (Table 16) represent real mechanisms — mental health affecting physical morbidity through neuroendocrine and behavioural pathways; parental economic resources

shaping children's educational opportunities — whose omission is a framework constraint rather than a denial that they exist. Cross-sectional DAG analysis cannot capture feedback dynamics that a longitudinal or dynamic-causal treatment would accommodate, and the directional choices reflect the identification demands of the HEE–MH estimand rather than a complete account of the temporal structure.

Fourth, no DAG can enumerate every possible common cause of exposure and outcome. Domain knowledge is incomplete, and structures not considered during construction may constitute omitted confounders that $\mathcal{A}$ does not address. FCI's finding that most recovered edges require a latent common cause is a direct empirical reminder of this. A further concern for area-level analysis is positivity: the joint distribution of $\mathcal{A}$ across the LSOA stock contains regions where HEE varies little — pre-1919 rural off-gas-grid solid-walled stock, where dwelling age, climate, and urbanicity jointly constrain the observed range of EPC ratings. Where treatment and control states are not both represented, an estimator extrapolates rather than contrasts. This motivates the flexible doubly-robust estimators of the companion analysis, which diagnose and partially correct overlap violations that linear regression averages over.

Four further limitations were raised by the reviewing panel and are recorded here because none is resolved by the graph.

The first concerns the exposure measure. An Energy Performance Certificate is issued when a dwelling is sold or newly let, so the certified stock of an area is not a random sample of its dwellings: a property retrofitted but not subsequently transacted does not appear, and areas differ in transaction rates. Identification therefore requires the additional assumption that this coverage bias operates comparably across areas, so that it does not correlate with the exposure contrast of interest. The assumption is not testable with these data and is stated rather than defended. It bears directly on the change-based design of the companion paper, since differential coverage is one route by which an area's apparent efficiency gain could reflect transaction activity rather than fabric improvement.

The second concerns the sign of the proximal mediator. The graph draws a single unsigned edge from HEE to *IEQ*, but the direction of that effect is contingent on how efficiency is achieved: a fabric measure installed with adequate purpose-provided ventilation improves indoor conditions, whereas one that increases airtightness without it can raise pollutant and moisture concentrations. An EPC rating records neither. The affordability channel through *Fuel_Poverty* is not exposed in this way, since lower running costs act in one direction. The consequence is that the two channels the graph identifies are not equally well behaved, and an average effect estimated across a stock of unknown installation quality conceals heterogeneity that the graph cannot represent.

The third concerns the scale on which the exposure is measured. Treating the SAP score as cardinal assumes that a given increment carries the same meaning at every point of the range, and treating the exposure–outcome relationship as linear assumes the same of the dose–response. Neither is established. The companion paper addresses the second by varying the functional form of the treatment across its specification grid; the first remains an assumption of the measure itself.

The fourth concerns the classification of *Tenure*. It is placed here among the structural attributes of the dwelling, and review argued that it is better understood as a characteristic of the occupants, who move between tenures. Identification is unaffected, since the construct enters $\mathcal{A}$ under either reading, but the mechanism the graph invokes — the landlord–tenant split incentive that suppresses retrofit — is specific to the private rented sector, while the area-level measure available here combines private and social renting. Social housing has in places been retrofitted at higher rates through public investment, which acts in the opposite direction, so the combined measure may attenuate the relationship it is intended to capture.

These limits apply to any observational causal analysis. What this paper establishes is prior to an effect estimate: a declared structure, a minimal adjustment set derived from it, and an empirical check that the structure is not contradicted by the data on which it will be used. The companion paper takes the adjustment set forward, moving from cross-sectional levels to change-based outcomes to remove time-invariant area heterogeneity, and estimating the effect with doubly-robust machine learning (**??**). The identification questions this paper leaves open — how baseline differences between areas are handled, and whether the recovered effect survives that choice — are settled there, not here.

6. Conclusions

Causal claims about home energy efficiency and mental health have outrun the structural assumptions needed to support them. This paper supplies those assumptions in a form that can be inspected, criticised, and tested. We construct an expert-informed directed acyclic graph of 25 constructs and 89 edges linking home energy efficiency to mental health at LSOA level, built from a synthesis of 53 peer-reviewed publications and revised through a three-stage expert review — asynchronous comment, a recorded workshop, and a sign-off round — with every edge, every exclusion and every review decision documented. Applying the backdoor criterion returns a minimal sufficient adjustment set of five constructs: Climate, Economic Capital, Housing Characteristics, Tenure, and Urbanicity. Education and demographic composition, which a conventional specification would adjust for, are excluded because their effects on the exposure are routed through those five.

Confronting the graph with area-level data for 31,802 English neighbourhoods leaves its central structural claim intact. Energy efficiency and mental health are non-adjacent in the expert graph and in every graph recovered by PC-stable and FCI, at three significance thresholds and under two outcome definitions. Encoding the expert causal ordering improves orientation substantially while leaving the skeleton unchanged, which indicates that domain knowledge supplies what these data cannot. FCI marks 28 of 39 edges as attributable to unmeasured common causes, including every edge incident to energy efficiency, and 77% of the graph's failed independence implications involve pairs it connects through an unmeasured node. Area-level associations in this setting are dominated by shared latent structure, which is the reason the graph carries latent nodes and an adjustment set at all.

Two results set the boundaries of what was tested. The disagreements not explained by latent structure concentrate on climate, whose winter-temperature proxy varies smoothly with geography and correlates with most area characteristics for reasons the graph does not model; resolving them requires a better measure and an explicit spatial model. And fuel poverty, the only mediator on the pathway with an area-level measure, shows a strong recovered link from energy efficiency but no residual association with mental health once area deprivation is adjusted for, indicating that ecological cross-sections are poorly suited to detecting mediation of this kind.

The contribution is an identification strategy rather than an effect estimate. Discovery corroborates the graph without establishing edge directions, which cross-sectional area-level data cannot do. The adjustment set derived here is taken forward in a companion paper that estimates the effect using change-based outcomes and doubly-robust machine learning.

CRediT authorship contribution statement

Chun Sang Au Yong: Conceptualisation, Methodology, Software, Validation, Formal Analysis, Investigation, Resources, Data Curation, Writing - Original Draft, Writing - Review & Editing, Visualisation, Project Administration. **Phil Symonds:** Writing - Review & Editing, Supervision. **Giorgos Petrou:** Writing - Review & Editing, Supervision.

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Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Claude 3.6 Sonnet and Opus developed by Anthropic in order to improve readability and language. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

A. Edge-by-Edge Evidence for the Causal DAG

Tables 6–15 present the full evidence base for each directed edge in the DAG. For each edge, the table provides a brief causal rationale and one or more supporting references. Edges are grouped thematically to align with the narrative review in Section 4.1.

Table 6: Edges into Home Energy Efficiency (HEE).

Edge	Rationale and supporting evidence
Economic_Capital → HEE	Household income constrains the capacity to invest in retrofit measures, which typically require substantial upfront capital even where government subsidies are available. Evidence from a controlled before-and-after study in low-income Welsh neighbourhoods demonstrates that energy efficiency investments are concentrated in areas targeted by income-based programmes (**?**). Quasi-experimental analysis of low-income English households shows that modest income increases translate directly into higher energy spending, consistent with a binding budget constraint on household energy behaviour (**?**).
Urbanicity → HEE	Rural dwellings in England have systematically lower EPC ratings than urban dwellings (median 59 versus 68), a gap only partially accounted for by housing stock composition. Analysis of 14 million English EPC records identifies rural-urban status as the single most important geographic predictor of dwelling-level energy efficiency, driven by older detached stock, absence of mains gas connection, and remoteness from retrofit services (**??**). Modern high-density urban construction complying with updated building regulations further widens this gap for new-build stock.
Policy_Scheme_Access → HEE	Household- and area-level uptake of government energy efficiency programmes directly determines whether retrofit measures are installed. The Energy Company Obligation (ECO, 2013–present, currently in its fourth iteration ECO4 running 2022–2026) has accounted for between 71% and 86% of measures installed under government schemes in each year since 2013, and for approximately 75% of measures installed in 2024 (**?**). The now-closed Green Deal (2013–2015) (**?**) and the ongoing Warm Home Discount scheme further contribute to the distribution of HEE through means-tested and property-targeted eligibility criteria (**?**).

Table 6 continued from previous page

Edge	Rationale and supporting evidence
Housing_Characteristics → HEE	Physical dwelling form (age, construction era, built form, and materials) is the strongest single determinant of baseline energy efficiency. Pre-1919 English properties average EPC 60, compared with 74 for post-1990 builds (**?**). Solid-walled properties, listed buildings, and detached houses with high surface-to-volume ratios face both higher baseline heat loss and greater technical constraints on retrofit feasibility (**??**).
Tenure → HEE	The landlord-tenant split incentive suppresses retrofit investment in the private rented sector: landlords bear the capital cost of improvements, while tenants capture the resulting bill savings, creating a principal-agent problem that depresses efficiency investment. Empirical evidence using US Residential Energy Consumption Survey data documents landlord underinvestment across multiple categories of residential energy efficiency, including space-heating systems, water-heating, window thickness, insulation, and weatherisation (**?**). These findings are corroborated in the residential rental market by **?** and **?**.
Climate → HEE	Regional climate shapes domestic heat demand profiles and thereby the marginal value of efficiency measures: longitudinal heat demand monitoring of UK social housing shows substantial spatial variation in heating demand reflecting regional weather conditions (**?**). Because the absolute energy and cost savings delivered by a given fabric measure scale with underlying heat demand, this variation alters the cost-effectiveness and geographic targeting of retrofit, and hence the efficiency state ultimately achieved across the stock.

Table 7: Edges into Mental Health (MH).

Edge	Rationale and supporting evidence
Education → MH	Higher educational attainment is associated with improved coping, health literacy, and self-efficacy, and is consistently linked to better adult mental health outcomes. Sibling fixed-effects analyses of UK Biobank data find that higher educational attainment remains significantly associated with fewer depressive symptoms after controlling for unobserved family-level confounders and polygenic scores, though the strength of these associations is reduced by 40–70% relative to standard OLS estimates once sibling controls are applied (**?**). Population-level evidence places education within the framework of fundamental social causation of mental health (**??**).
Economic_Capital → MH	Poverty, unemployment, and financial stress are established causal risk factors for common mental disorders. Using panel data from the UK Household Longitudinal Study, **?** estimate that falling below the poverty line increases common mental disorder prevalence by 2.5 percentage points. This finding is corroborated by systematic review and meta-analytic evidence synthesising causal studies of poverty and mental health in low- and middle-income as well as high-income contexts (**??**).

Table 7 continued from previous page

Edge	Rationale and supporting evidence
Tenure → MH	Housing tenure insecurity, particularly in the private rented sector, generates anxiety through fear of eviction, uncontrolled rent increases, and limited control over dwelling conditions. Longitudinal analysis in the UK and international contexts demonstrates that housing insecurity independently predicts psychological distress, poor sleep, and hypertension, beyond the effect of income or housing physical quality (**???**).
Overcrowding → MH	Residential overcrowding generates psychological distress through reduced privacy, interpersonal conflict, constrained personal space, and reduced control over the domestic environment. Longitudinal analysis of Chilean household trajectories finds that sustained overcrowding increases the probability of poor mental health, with effects attenuated but not eliminated after accounting for income (**?**). Parallel evidence from Sweden and New Zealand confirms the association in migrant and general populations (**??**).
Climate → MH	Extreme weather events — particularly flooding and heatwaves — directly cause psychological trauma. A UK-focused meta-analysis of post-extreme-event mental health outcomes reports 12-month prevalence of 30% for PTSD, 21% for depression, and 20% for anxiety among those directly affected (**?**). This edge captures the direct mental health consequences of climate exposure, distinct from indirect pathways through housing degradation and IEQ.
Air_Pollution → MH	Chronic exposure to ambient $PM_{2.5}$ and NO_2 is associated with increased depression and anxiety risk, proposed to operate through neuroinflammation and oxidative stress. A systematic review and meta-analysis identifies a pooled association of 10–18% higher depression risk per 10 µg/m^3 $PM_{2.5}$ (**?**), corroborated in an updated synthesis by **?**, while **?** provides a contemporary review of the evidence base and mechanistic hypotheses.
Green_Space → MH	Access to and quality of green and blue space is protective against depression and anxiety, with proposed mechanisms including attention restoration and chronic stress reduction. A ten-year longitudinal dynamic panel study of 2.3 million adults in Wales finds significant protective associations between ambient greenness, local green space access, and subsequent mental health (**?**). Twin analyses and a recent meta-analysis further support the association (**???**).
Noise → MH	Chronic environmental noise exposure — primarily from road, rail, and aircraft traffic — is hypothesised to affect mental health through annoyance, sleep disturbance, and sustained physiological stress. Evidence quality is mixed: a meta-analysis of road traffic noise and depression found a pooled 4% increase in depression odds per 10 dB(A) L_{den} and 12% for anxiety, though both estimates had confidence intervals crossing zero and the synthesis itself graded the underlying evidence as "very low" quality (**?**). A more recent meta-analysis finds statistically significant effects specifically for aircraft noise (12% increase in depression risk per 10 dB L_{den}) but weaker, non-significant effects for road and rail traffic sources (**?**). **?** offers a contemporary review integrating mechanistic evidence.

Table 7 continued from previous page

Edge	Rationale and supporting evidence
IEQ → MH	Indoor damp, mould, cold, and indoor air pollutants are associated with distress, perceived loss of control, and measurable increases in depression and anxiety. A UKHSA scoping review finds that 88% of included studies report significant associations between dampness/mould exposure and psychological outcomes including depression, anxiety, and perceived loss of control over the home environment (**?**). **?** establishes the dampness–depression pathway operates both through mould-related physical illness and through perceived control; **?** provides a framework for the cold-housing/MH mechanism.
Social_Capital → MH	Area-level social capital — community trust, civic participation, and associational density — is inversely associated with common mental disorder prevalence through pathways of social support, collective efficacy, and sense of belonging. A systematic review of cognitive social capital and mental disorder finds consistent protective associations across multiple study designs and contexts (**?**). Updated evidence confirms this across recent UK and international studies (**??**).
Isolation → MH	Social isolation and loneliness are robust, independent risk factors for depression, anxiety, and cognitive decline. An umbrella review of systematic reviews finds that loneliness more than doubles the odds of developing new-onset depression across longitudinal studies (**?**), with **?** providing a contemporary synthesis of the evidence and mechanistic pathways.
Accessibility → MH	Physical access to healthcare, employment, and social services affects mental health both directly (through increased stress and reduced ability to obtain treatment) and indirectly (through participation in economic and social life). Evidence from UK health service inequalities research and international accessibility studies supports the association (**??**).
Digital_Capital → MH	Digital capital — broadband access, device availability, and digital literacy — has a net protective effect on mental health at the population level, primarily through access to mental health resources, telehealth, and maintenance of social connection. Digital exclusion disproportionately affects people with severe mental illness and older adults in deprived communities, compounding existing mental health inequalities (**??**). Despite adverse effects of digital overuse, at the LSOA aggregate level relevant to the present DAG, the access-to-services channel dominates.
Substance_Abuse → MH	Substance use disorders are highly comorbid with mental health disorders. A meta-analysis of epidemiological surveys finds that drug use disorder is associated with 3.8-fold increased odds of major depression, and alcohol use disorder with approximately 2.4-fold increased odds (**?**).
PH → MH	Chronic physical illness increases depression risk through pathways of disability, pain, inflammation, and reduced quality of life. A mediation analysis using UK longitudinal data demonstrates that physical health substantially mediates the income-mental-health relationship (**?**); a comprehensive recent review synthesises the comorbidity patterns between major depressive disorder and physical disease (**?**). The population-level burden is described in **?**.

Table 7 continued from previous page

Edge	Rationale and supporting evidence
Neighbourhood_Stress → MH	Chronic neighbourhood stressors — principally crime and perceived unsafety — generate fear, hypervigilance, and sustained psychological distress. A meta-analysis of 63 studies finds significant pooled associations between neighbourhood crime and depression, anxiety, and general distress (**?**). Non-crime dimensions of neighbourhood stress (physical disorder, antisocial behaviour) are included as part of the construct theoretically, though the dominant empirical evidence base operationalises neighbourhood stress through crime exposure.
Demographics → MH	Age, gender, ethnicity, and household composition shape population-level distributions of common mental disorder. Women have substantially higher CMD prevalence than men in England (approximately 24% versus 15% in recent APMS data); ethnic minorities face discrimination-related and migration-related mental health stressors (**??**). Fundamental-cause framing places these patterns within the broader social determinants of mental health (**?**).

Table 8: Outgoing edges from Education and Economic Capital.

Edge	Rationale and supporting evidence
Education outgoing edges	
Education → Economic_Capital	Educational attainment causally affects adult earnings through labour market returns. UK-based instrumental variable estimates exploiting the 1947, 1963, and 1972 compulsory schooling reforms provide a robust case for approximately a 6% rate of return to education for men (**?**). Broader international reviews of the returns-to-schooling literature report returns in the range of 5–10% per additional year of schooling across developed economies, with UK-specific estimates clustering at the lower end of this range once reform-based identification strategies are applied (**??**). This edge also serves as the pathway through which education indirectly affects home energy efficiency, replacing the direct *Education* → *HEE* edge excluded in Section 4.1.8.
Education → PH	Education is associated with reduced all-cause morbidity and mortality through pathways including health literacy, health behaviours, occupational exposures, and access to healthcare. A global systematic review and meta-analysis of 603 studies reports a reduction of approximately 1.9% in adult mortality risk per additional year of schooling, with effects concentrated in causes of death amenable to health-behavioural mediation (**?**).
Economic Capital outgoing edges	
Economic_Capital → PH	Socioeconomic status operates as a fundamental cause of physical health disparities, embodying substitutable resources — income, wealth, social connections — that protect health through a shifting set of proximate mechanisms (**??**). The US literature on the urban-rural dimension of this relationship is characterised in **?**; the pattern is corroborated in UK data through the Marmot Review evidence base.

Table 8 continued from previous page

Edge	Rationale and supporting evidence
Economic_Capital → Housing_Characteristics	Household income and wealth determine the quality of accessible housing stock. Using Family Resources Survey and English Housing Survey data, **?** show that households in the lowest UK income quintile are substantially more likely to live in homes that are hazardous, difficult to heat, or in poor repair, and that the shrinking pool of properties affordable to housing benefit recipients is of systematically lower quality than the UK housing stock average. **?** document similar stratification in the US context.
Economic_Capital → Tenure	Wealth and income determine access to owner-occupation in the UK housing market. Social renters in the UK have an after-housing-cost poverty rate of 46%, private renters 34%, and mortgagors 11%, with outright owners at 15% (**?**).
Economic_Capital → Overcrowding	Lower household income constrains consumption of housing space relative to household size. More than two million children from 746,000 families in England live in overcrowded conditions, overwhelmingly concentrated in low-income households (**?**).
Economic_Capital → Policy_Scheme_Access	Means-tested UK energy assistance programmes — most notably Warm Home Discount Core Group 2 — condition eligibility on receipt of qualifying income-related benefits, creating a direct link between area-level economic capital and the reach of energy policy instruments (**??**). Expert review proposed reversing this edge (*Policy_Scheme_Access* → *Economic_Capital*, on the basis that scheme receipt relieves household budgets); the *Economic_Capital* → *Policy_Scheme_Access* orientation is retained because eligibility is conditioned on pre-existing income, but the orientation is flagged for an orientation-sensitivity analysis in which the edge is reversed to confirm that the target estimand is robust to this directional choice. That analysis is reported in Table 17: reversal forces *Policy_Scheme_Access* into the adjustment set as an unmeasured confounder, so the committed orientation is retained as a defended, bounded choice.
Economic_Capital → Neighbourhood	Household economic resources determine residential neighbourhood through housing market sorting. UK evidence on ethnic and income-based neighbourhood selection demonstrates that higher economic resources facilitate movement to more advantaged neighbourhoods, while residents of constrained means experience limited locational choice (**??**).
Economic_Capital → Social_Capital	Economic resources facilitate civic participation through time, financial capacity, and psychosocial stability. A UK analysis of social capital gradients finds a participatory gap between lower- and higher-income groups, though with substantial residual heterogeneity — deprived communities can and do maintain strong social capital through alternative mechanisms, including kinship networks and informal mutual aid (**?**). A meta-analysis of socioeconomic gradients in civic engagement corroborates this pattern across high-income countries (**?**).

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Edge		Rationale and supporting evidence
Economic_Capital Digital_Capital	→	Income determines broadband access, device ownership, and digital literacy. Ofcom's UK digital exclusion evidence finds a strong income gradient in internet use, with approximately one in five adults in households earning under £25,000 never using the internet (**?**). Broader evidence from European and US studies confirms the socioeconomic stratification of digital access and the emergence of "relative digital deprivation" as a secondary inequality (**?**).
Economic_Capital Substance_Abuse	→	Economic deprivation increases the risk of problem substance use. Alcohol-specific death rates in the most deprived decile of UK local authorities are approximately 4.2 times higher than in the least deprived decile (**?**); the broader socioeconomic patterning of alcohol-related harm is reviewed in **?**.

Table 9: Outgoing and incoming edges for the Fuel Poverty mediator.

Edge	Rationale and supporting evidence
Inbound edges to Fuel Poverty	
HEE → FP	The energy efficiency of the dwelling stock is a definitional determinant of fuel poverty: the Low Income Low Energy Efficiency (LILEE) metric used in England embeds the efficiency rating directly in its threshold, so areas with less efficient stock mechanically exhibit higher modelled prevalence (**?**). Beyond the definitional link, the mechanism is that a more efficient dwelling requires less delivered energy to reach an adequate internal temperature, lowering the expenditure needed for thermal comfort; **?** reports that home energy-efficiency schemes significantly reduce fuel-poverty prevalence. The edge is oriented HEE → FP because dwelling efficiency is logically prior to the affordability outcome it helps determine (**?**).
EC → FP	Household income is the second definitional input to fuel-poverty: the low-income component is assessed against an income threshold, so area-level economic capital is constitutive of measured fuel poverty independent of dwelling efficiency (**?**). The mechanism is direct—for a given energy cost, lower income raises the share of expenditure absorbed by energy and the likelihood of falling below the affordability threshold (**?**). This edge is required for FP to be correctly specified; omitting it would misrepresent fuel poverty as a function of efficiency alone. EC already enters the graph as an upstream confounder of HEE (EC → HEE); the EC → FP edge is conceptually distinct, capturing income's role in affordability *after* efficiency is determined, and the two should not be conflated (**?**).

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Edge	Rationale and supporting evidence
C → FP	Climatic heating demand is a recognised driver of fuel poverty: colder areas require more delivered energy to maintain an adequate internal temperature, raising the energy cost faced for a given dwelling efficiency and income. This edge parallels the existing C → HEE and C → IEQ edges, under which climate already operates through heating demand. The spatial patterning of fuel poverty in England is documented by **?**. Energy *price* is excluded as a separate determinant because, in the England-and-Wales LSOA cross-section, tariffs are effectively national and do not vary across units, contributing nothing to identification.
Outbound edges from Fuel Poverty	
FP → MH	Fuel poverty is consistently associated with poorer mental health: in a global scoping review, **?** found a detrimental association in 46 of 47 included studies and organised the pathways into economic, social, behavioural, and environmental mechanisms. The association is observational and confounded by income and housing quality, so it does not by itself establish causation; here those common causes (EC, H) are conditioned, and the residual edge carries the affordability mechanism net of them. The proposed mechanism is psychosocial financial strain—anxiety over unaffordable bills, the heat-or-eat trade-off, and the chronic stress of an under-heated home (**??**). This edge constitutes the economic/affordability arm of HEE's effect on mental health, running parallel to the physical/thermal arm carried by HEE → IEQ → MH.
FP → PH	Fuel poverty is linked to adverse physical health, particularly cold-related cardiovascular and respiratory morbidity and excess winter mortality (**??**). As with FP → MH, the raw association is confounded by income and dwelling characteristics, which the adjustment set addresses; the edge represents the effect of unaffordable energy net of those common causes. The mechanism operates through sustained exposure to inadequate indoor temperatures and the diversion of constrained budgets away from food and health-protective expenditure, with energy-poor households forgoing medicine and healthcare to meet bills (**?**). Evidence is not uniform—**?** found no clear reduction in emergency hospital admissions following efficiency investment—so the causal claim is made cautiously. Because PH → MH already exists, this edge also opens an indirect FP → PH → MH sub-path.

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Edge	Rationale and supporting evidence
FP → IEQ	Fuel poverty degrades indoor environmental quality through a behavioural rationing mechanism that is distinct from the physical-fabric pathway carried by HEE → IEQ. Households unable to afford adequate warmth ration heating—under-heating rooms, heating intermittently, or restricting heating to a single room—which lowers internal surface temperatures, and, combined with everyday moisture generation, promotes condensation below the dew point, persistent damp, and mould growth. **?** find that fuel poverty increases the risk of mould contamination in social housing independently of occupant risk perception and ventilation behaviour, and **?** document the broader cold-housing pathway. The edge is oriented FP → IEQ because affordability-driven heating behaviour is logically prior to the indoor conditions it produces. It is conditionally distinct from the dwelling-form channel (*Housing_ Characteristics* → *IEQ*), the occupancy-load channel (*Overcrowding* → *IEQ*), and the climate heating-demand channel (*Climate* → *IEQ*), and it provides a second route—alongside the existing FP → MH and FP → PH edges—through which affordability reaches health, namely FP → IEQ → {MH, PH}.

Table 10: Outgoing edges from Urbanicity.

Edge	Rationale and supporting evidence
Urbanicity → Neighbourhood	Urban–rural classification shapes neighbourhood density, land use, infrastructure provision, and the distribution of local amenities and disamenities. Although many wealthy urban neighbourhoods exist and rural areas exhibit substantial socioeconomic heterogeneity, the urban-rural dimension operates as a structural determinant of neighbourhood composition that sits upstream of deprivation rather than as a simple synonym for it. Evidence of the persistence of area-level urban and deprivation structures in England and Wales over five decades ($r = 0.706$ between 1971 and 2021 deprivation) documents this structural role (**?**).
Urbanicity → Policy_Scheme_Access	Energy policy instruments and household uptake of energy assistance schemes are differentially distributed across urban and rural areas. Rural fuel poverty exhibits distinct characteristics — notably concentration among households off the gas grid and in pre-1919 solid-walled properties — which both shape eligibility for specific retrofit programmes and create barriers to uptake in remote areas (**?**).

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Edge	Rationale and supporting evidence
Urbanicity → Isolation	Urbanicity affects the geographic and logistical substrate of social isolation, distinct from (though correlated with) the relational pathways captured by *Social_Capital*. Rural residents face greater physical distances from family, services, and community venues, while urban residents — despite higher population density — report elevated loneliness attributable to weaker neighbourly ties and transient residency patterns (**???**). The distinction from *Social_Capital* is that isolation here refers to individual-level geographic and physical access to social contact, whereas social capital captures area-level civic participation and associational density; the two are empirically correlated but represent distinct theoretical constructs.
Urbanicity → Economic_Capital	The UK urban–rural pattern in economic capital is heterogeneous and depends on the specific rural sub-category and the measure of income used. Analysis of British Household Panel Survey data finds that urban wages are significantly lower than accessible rural wages but significantly higher than remote rural wages, even after adjusting for observed worker characteristics (**?**). Urban wage gains are further partially offset by higher urban housing costs and cost-of-living differentials, particularly in London and the South East, such that the net effect on household disposable income after housing costs is ambiguous and area-specific (**?**). The edge is retained to reflect the systematic *structural* relationship between urbanicity and labour market access, while acknowledging that its net magnitude in real terms varies substantially across the urban–rural classification.
Urbanicity → Education	Socioeconomic disadvantage interacts with rurality in determining educational outcomes. On average, rural pupils slightly outperform urban peers at GCSE in England, but this masks underperformance among rural pupils from deprived backgrounds: once income deprivation is controlled for, rural pupils from the most deprived areas underperform their urban counterparts by approximately 2 percentage points on the 5EM GCSE measure on average, rising to 4–8 percentage points at the extreme ends of deprivation. This pattern is attributed to transport barriers, school sparsity, and narrower post-16 educational choice in rural areas (**??**).
Urbanicity → Social_Capital	Rural areas exhibit systematically higher levels of bonding social capital — trust, associational density, and neighbourly contact — than urban areas in England. ONS social capital statistics report that approximately 82% of rural residents talk regularly to their neighbours compared with 72% of urban residents (**?**). The mechanism operates through population stability, smaller social units, and the functional necessity of mutual assistance in low-density contexts.
Urbanicity → Digital_Capital	A pronounced urban–rural digital divide persists in the UK: over 53% of deep-rural areas cannot achieve adequate broadband speeds, compared with approximately 5% in urban areas. This gap reflects the geographic economics of broadband infrastructure deployment and is a structural feature that direct investment has only partially addressed (**?**).

Table 11: Outgoing edges from U_AreaHistory (latent).

Edge	Rationale and supporting evidence
U_AreaHistory → Neighbourhood	Historical planning decisions — slum clearance, council estate construction, industrial siting, motorway routing — physically produced the spatial structure within which present-day neighbourhoods are situated. These decisions shape the housing stock, transport infrastructure, and land-use mix that define neighbourhood boundaries and conditions today, even where present-day deprivation levels do not directly reflect historical patterns. The degree of persistence in area-level structures is substantial: deprivation rankings of areas in England and Wales in 1971 correlate at $r = 0.706$ with deprivation rankings in 2021, demonstrating remarkable stability of relative spatial outcomes over half a century (**??**).
U_AreaHistory → Economic_Capital	The legacy of twentieth-century deindustrialisation — colliery closures, steel and manufacturing decline, dock closures — produced persistent intergenerational economic deprivation in affected communities. Causal-inference analysis of UK colliery closures finds that children exposed to pit closures during formative years experienced systematically worse labour-market outcomes throughout adult life, with effects persisting into subsequent generations (**?**). The broader structural economic disadvantage of former industrial "left-behind" areas is documented by **?**.
U_AreaHistory → Education	Historical underinvestment in schools serving deprived areas perpetuates educational disadvantage across generations. UK cohort analyses show that between 1981 and the late 1990s, university graduation rates among young people from the poorest 20% of families rose by only 3 percentage points, compared to a 26 percentage point rise for those from the richest 20% — despite substantial overall expansion of higher education during the same period (**?**). Evidence of the persistence of local-authority-level spending inequalities in English schools into the 2020s is provided in **?**.
U_AreaHistory → PH	Former industrial and deindustrialised "left-behind" areas exhibit systematically worse health outcomes, including substantially reduced life expectancy. In the UK's left-behind neighbourhoods, men live on average 3.7 years fewer and women 3.0 years fewer than national averages (**?**). The US-based redlining literature provides complementary evidence that historical area-level discrimination exerts long-run health effects operating partly through material deprivation and partly through unmeasured structural channels (**?**).
U_AreaHistory → Tenure	The Right to Buy scheme, introduced by the Housing Act 1980, gave eligible council tenants in England and Wales the statutory right to purchase their council home at substantial discount. The policy reshaped UK tenure composition at area level: England's social housing stock fell from approximately 5.5 million units in 1980 to 4.1 million in 2021, with council housing specifically experiencing the largest decline. Cumulative sales over 1980–2023 exceed two million council homes, with sales rates varying sharply across regions and over time in response to discount policy changes (**???**). The tenure landscape produced by these sales — a reduced council sector, expanded owner-occupation, and latterly a substantial private-rental sector composed of former council homes — reflects a historical policy decision rather than a contemporary market outcome.

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Edge	Rationale and supporting evidence
U_AreaHistory → Social_Capital	Post-war slum clearance and population relocation disrupted established kinship and community networks in UK cities, with documented effects on the social fabric of receiving neighbourhoods and on the capacity for collective action in displaced communities (**??**). Subsequent deindustrialisation further fragmented community bonds in former mining and manufacturing areas, where occupational solidarity had served as a key substrate for civic participation.

Table 12: Outgoing edges from Neighbourhood (scale-bridging construct).

Edge	Rationale and supporting evidence
Neighbourhood → Air_Pollution	Ambient air pollution varies systematically across areas of differing deprivation in England. Recent LSOA-level analysis covering 2003–2023 finds that the most deprived 20% of English LSOAs experienced 8% higher average $PM_{2.5}$ concentrations than the least deprived quintile in 2023, a gap that has widened since 2017 (**?**). Sectoral-emissions analysis at LSOA level finds that all 24 minoritised ethnic groups studied face higher local NOx and $PM_{2.5}$ emissions than socioeconomically matched majority-group populations (**?**), consistent with longer-standing evidence from **?** and **?**. These findings demonstrate robust spatial correlation between area-level disadvantage and pollution exposure, but do not on their own establish a causal pathway at the area level. A causal interpretation of the scale-bridging edge nonetheless remains well-supported by quasi-experimental evaluations of UK traffic and low-emission interventions: the 2019 London Ultra Low Emission Zone produced a 19.6% reduction in NO_2 at traffic sites and 8.2% at urban background sites using augmented synthetic control identification (**?**); the Birmingham Clean Air Zone achieved 3.4–6.6% NO_2 reductions at roadside sites using the same methodology (**?**); and evaluations of London Low Traffic Neighbourhoods using difference-in-differences identification found 5.7% NO_2 reductions within intervention areas (**?**). These quasi-experimental findings establish that area-level traffic and planning decisions causally determine local pollution exposure, providing mechanism-level support for the scale-bridging claim that upstream factors aggregate through Neighbourhood to shape Air_Pollution.
Neighbourhood → Green_Space	Green and blue space provision is a feature of the local area determined by planning decisions, land-use patterns, and historical investment in public amenities — all of which aggregate upstream factors into an area-level environmental exposure. A recent UK study of 6,791 English MSOAs finds that grassland is inequitably distributed across urban neighbourhoods by Index of Multiple Deprivation: among the most deprived quintile of urban MSOAs in England, a 1% increase in grassland area is associated with a 37% reduction in annual preventable deaths (incidence rate ratio 0.63, 95% CI 0.52–0.76), with the authors explicitly cautioning that this is an observational estimate subject to ecological-inference concerns rather than a causal effect (**?**). International evidence of unequal green space distribution is reviewed in **?** and **?**.

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Edge	Rationale and supporting evidence
Neighbourhood → Accessibility	Physical access to healthcare, employment, and services is determined by the spatial distribution of transport networks and service locations at the area level, not by individual household characteristics. The scale-bridging role captures how upstream factors aggregate into area-level accessibility provision. Evidence of the resulting access burden is documented in UK transport research: low-income households in poorly connected areas spend approximately 25% of their disposable income on commuting costs, compared to an average of around 13%, reflecting constrained mobility choices that follow from area-level transport provision rather than individual household preference (**?**). Complementary spatial-accessibility evidence is provided by **?**.
Neighbourhood → Neighbourhood_Stress	Neighbourhood stressors — crime, disorder, anti-social behaviour, and dereliction — cluster spatially as area-level exposures. Their distribution reflects the aggregation of upstream economic and historical factors into place-based experiences of unsafety and disorder. In England, 25% of people in the most deprived decile live in the top 10% highest-crime areas, compared with 3.1% of those in the least deprived decile (**?**). Corroborating evidence on neighbourhood disorder is provided in **?**.
Neighbourhood → Housing_Characteristics	Area-level conditions are associated with systematic differences in the physical housing stock. Overcrowded properties cluster sharply in the most deprived 20% of LSOAs in England (**?**), and affordable properties in the UK are 15% more likely to have an EPC rating of D or below than the national housing stock average (**?**). These patterns are cross-sectional and do not on their own establish that area-level aggregation causally produces housing-stock differences. A causal interpretation of the scale-bridging edge rests on the mechanism of ongoing maintenance and investment: the rate at which dwellings are maintained, upgraded, or replaced is shaped by neighbourhood-level rental markets, landlord portfolios, local authority investment, and regeneration decisions that operate at area scale and aggregate upstream household-level investment capacity. Quasi-experimental evidence on this specific mechanism at UK area level is limited, and the edge is retained on the basis of plausibility and theoretical consistency with the scale-bridging framework rather than on direct causal demonstration.
Neighbourhood → Noise	Exposure to environmental noise — from traffic, rail, aircraft, and industrial sources — is an inherently area-level phenomenon determined by the spatial geometry of transport and industrial infrastructure. The scale-bridging role of Neighbourhood captures how upstream land-use and planning decisions aggregate into differential area-level noise exposure. Social inequalities in transport noise are observed across English urban areas, with the most deprived neighbourhoods facing systematically higher exposure (**??**).

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Edge	Rationale and supporting evidence
Neighbourhood → Social_Capital	Social capital is an intrinsically area-level phenomenon: trust, associational density, and civic participation are properties of a community rather than of an individual. The edge from Neighbourhood captures how upstream factors — economic, historical, and structural — aggregate into the area-level context within which social capital forms or fails to form. UK-based nationally representative analysis finds that material deprivation, rather than ethnic or demographic diversity, is the principal neighbourhood-level predictor of reduced social capital (**??**), consistent with the aggregation mechanism operating through upstream economic rather than demographic channels.

Table 13: Outgoing edges from Neighbourhood_Stress (excluding Neighbourhood_Stress → MH; see Table 7).

Edge	Rationale and supporting evidence
Neighbourhood_Stress → PH	Chronic exposure to crime, disorder, and perceived unsafety operates on physical health through a sustained physiological stress response. The hypothesised mechanism is allostatic load: repeated activation of the hypothalamic–pituitary–adrenal axis and sympathetic nervous system in response to chronic stressors produces cumulative dysregulation across cardiovascular, metabolic, and immune systems, manifesting as elevated blood pressure, inflammatory markers, and metabolic risk factors. Multicohort evidence from the UK Whitehall II, Swiss CoLaus, and Portuguese EPIPorto studies documents that neighbourhood socioeconomic deprivation is associated with higher allostatic load even after adjustment for individual socioeconomic position (**?**). A systematic review and meta-analysis finds that high allostatic load is associated with a 22% increase in all-cause mortality and a 31% increase in cardiovascular-disease mortality (), linking the neighbourhood-stress-to-allostatic-load pathway to clinically meaningful PH outcomes. Complementary evidence from a systematic review of violence exposure and cardiovascular health identifies consistent associations between exposure to neighbourhood violence and cardiometabolic risk (**?**).
Neighbourhood_Stress → Isolation	Fear of crime produces social withdrawal as an individual adaptive strategy. Residents exposed to crime, perceived unsafety, or violence are more likely to restrict outings, avoid public spaces, and limit interaction with neighbours, reducing opportunities for social contact and increasing loneliness. Evidence from UK longitudinal studies links neighbourhood disorder and perceived unsafety to reduced social interaction and increased loneliness (**?**).

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Edge	Rationale and supporting evidence
Neighbourhood_Stress → Social_Capital	The mechanism operating through this edge is distinct from the aggregation channel represented by the *Neighbourhood* → *Social_Capital* edge in Table 12. Whereas that edge captures area-level material deprivation operating on social capital through community stability and civic infrastructure, the present edge operates through a fear-and-withdrawal mechanism: residents exposed to crime and disorder retreat into private space, reduce participation in local associations, and form fewer of the thin trust-based ties with neighbours that constitute social capital. The classic Chicago neighbourhood study of **?** establishes that concentrated disorder undermines collective efficacy independently of material deprivation, and UK-based evidence by **?** reproduces the finding that deprivation-and-disorder — not ethnic diversity — is the principal driver of reduced neighbourhood social cohesion.
Neighbourhood_Stress → Substance_Abuse	Neighbourhood stressors increase substance misuse through two complementary channels: a physiological stress pathway in which chronic HPA-axis activation drives self-medication behaviour (**?**), and a structural drug-market pathway in which high-crime and disorder-concentrated neighbourhoods exhibit denser local drug availability and lower perceived sanctions against use (**?**). Both channels contribute to higher substance-use prevalence in stressed neighbourhoods independently of individual socioeconomic position.

Table 14: Outgoing edges from Housing_Characteristics, Climate, Air_Pollution, Green_Space, and Noise (environmental mediators, excluding edges in Tables 6–7).

Edge	Rationale and supporting evidence
Housing_Characteristics outgoing edges	
Housing_Characteristics → PH	Poor housing is linked to respiratory disease, cardiovascular stress, and injuries through multiple pathways including structural hazards, inadequate thermal performance, and environmental exposures at the dwelling level. A systematic review identified 19 housing-related risk factors for respiratory disease (**?**), with corroborating evidence from **?** and **?** on the broader cardiovascular and health-injury burden of poor housing.
Housing_Characteristics → IEQ	Physical dwelling form — age, construction era, envelope materials, ventilation design, and maintenance condition — directly determines indoor temperature regulation, humidity, air quality, and mould risk. Systematic-review evidence identifies a broad range of dwelling-form attributes associated with IEQ outcomes (**???**).
Housing_Characteristics → Isolation	High-rise and poorly designed housing reduces opportunities for incidental social interaction and contributes to social withdrawal. A systematic review finds that living in high-rise housing is associated with worse social and mental health outcomes compared with lower-rise residential typologies (**?**).
Overcrowding outgoing edges (to IEQ)	

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Edge	Rationale and supporting evidence
Overcrowding → IEQ	Residential overcrowding degrades indoor environmental quality through elevated per-occupant moisture generation (respiration, cooking, bathing, laundry) and reduced effective ventilation per occupant, distributed over a fixed building envelope and ventilation capacity. This mechanism is distinct from the dwelling-form pathway captured in the *Housing_Characteristics* → *IEQ* edge: the envelope determines ventilation design, while occupancy determines the load placed on it. Empirical evidence links occupant density to elevated indoor humidity, condensation risk, and mould prevalence in UK and international housing (**???**). The English Housing Survey operationalises this link through the Housing Health and Safety Rating System (HHSRS) damp-and-mould hazard assessment, which incorporates occupancy density as a contributing factor alongside dwelling condition.
Climate outgoing edges	
Climate → Housing_Characteristics	Extreme weather — flooding, heat, and storms — degrades the structural integrity and condition of dwellings. In England, approximately 6.3 million properties are currently at flood risk, with projected increases by 2050 as climate change intensifies (**??**). Coastal and geographic exposure operates through the same channel and is, following expert review, represented within the exogenous *Climate* construct rather than as a separate node: dwellings in coastal and highly exposed LSOAs face elevated coastal-flood and storm-surge risk, wind-driven rain, and accelerated salt- and wind-weathering of the building fabric, which raise rates of fabric deterioration and repair need.
Climate → IEQ	Extreme outdoor temperatures propagate to indoor environments with substantial time lags and amplification dependent on dwelling fabric. Self-reported evidence from a national online survey of approximately 1,580 UK households conducted following the record 2022 heatwave, when outdoor temperatures exceeded 40°C for the first time in the UK, reports a rise in household-reported indoor overheating from 20% (2011) to 82% (2022) (**?**). These subjective figures are higher than those obtained from objective indoor-temperature monitoring: using the 2011 Energy Follow-Up Survey with CIBSE TM59 criteria, approximately 26% of English dwellings exceeded Criterion 2 (**?**), with higher prevalence projected under warming scenarios (**?**). Climate additionally drives indoor damp through moisture ingress: wind-driven (driving) rain penetrates the building envelope—disproportionately in exposed coastal and high-rainfall western areas and in solid-walled stock—raising wall moisture content, condensation risk, and mould, a mechanism recognised in UK building-fabric and exposure guidance (**?**). Flooding additionally introduces damp and mould, with parallel IEQ consequences for indoor air quality (**?**).
Air_Pollution outgoing edges	
Air_Pollution → PH	Long-term exposure to ambient air pollution causes respiratory and cardiovascular disease and premature mortality. A multi-country analysis of 854 European cities estimates that each 10 $\mu g/m^3$ increase in $PM_{2.5}$ is associated with approximately an 8% increase in all-cause mortality (**?**).

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Edge	Rationale and supporting evidence
Air_Pollution → IEQ	Outdoor particulate pollution infiltrates indoor environments through the building envelope, with the degree of infiltration dependent on airtightness, ventilation, and occupant behaviour. A systematic review of 77 studies covering over 4,000 homes (approximately 1,000 in the United States and 150 across Europe) reports a mean infiltration factor for $PM_{2.5}$ of 0.55, indicating that on average more than half of ambient $PM_{2.5}$ penetrates and persists indoors (**?**). UK-specific infiltration data is limited, though dwelling airtightness and ventilation patterns in the English housing stock suggest comparable infiltration behaviour (**?**).
Green_Space outgoing edges	
Green_Space → PH	Access to green space reduces cardiovascular-disease risk and all-cause mortality through pathways including physical activity, stress reduction, and heat mitigation. A meta-analysis synthesising evidence from over 100 million individuals reports a 2–3% reduction in cardiovascular-disease mortality per 0.1-unit increase in Normalised Difference Vegetation Index (NDVI), a standardised remote-sensing measure of vegetation density ranging from −1 (non-vegetated surfaces) to +1 (dense vegetation) (**???**). A 0.1-unit increase corresponds, for example, to the difference between low-density suburban vegetation and an area containing established parks or tree cover.
Green_Space → IEQ	Urban vegetation moderates the urban heat island effect, lowering ambient air temperatures by approximately 1–7°C depending on park size, vegetation type, and climate (**???**). The indirect effect on indoor temperatures operates through reduced outdoor-to-indoor heat transfer and is sensitive to building envelope and ventilation regime.
Green_Space → Air_Pollution	Urban trees and vegetation remove particulate matter through dry deposition and absorb gaseous pollutants through stomatal uptake. US urban trees are estimated to remove approximately 711,000 tonnes of pollutants annually (**??**).
Noise outgoing edges	
Noise → PH	Chronic exposure to environmental noise — principally from road, rail, and aircraft sources — causes cardiovascular disease through sustained sympathetic activation and disrupted sleep. A WHO-commissioned quantitative review of the environmental-noise evidence base finds consistent dose-response associations between community noise exposure and cardiovascular morbidity (**?**), with mechanism-level support from **?**. Evidence on occupational-noise exposure, which is distinct from the environmental-noise regime that characterises LSOA-level exposure, identifies approximately 1.81× higher hypertension risk at workplace noise levels $\geq$ 80 dB(A) (**?**), providing complementary mechanism-level evidence on the noise–cardiovascular pathway while noting the higher exposures typical of occupational settings.
Noise → IEQ	External noise penetrates indoor environments primarily through windows, doors, and building envelope weaknesses. UK building standard BS 8233:2014 specifies indoor noise targets to protect occupants from outdoor noise ingress (**??**).

Table 15: Groups 12–14 — HEE outgoing, IEQ outgoing, Social/Psychological Mediator, and Demographics outgoing edges.

Edge	Rationale and supporting evidence
HEE outgoing edges	
HEE → IEQ	Energy efficiency improvements alter indoor environments in complex ways. Properly implemented retrofits that incorporate adequate ventilation yield net IEQ gains; however, retrofits undertaken without ventilation provision can trap indoor pollutants and worsen air quality. Empirical evidence documents both the potential health benefits and the IEQ risks depending on whether ventilation is upgraded alongside envelope improvements (????).
HEE → Noise	Double- and triple-glazing, solid wall insulation, and improvements in airtightness reduce the transmission of external noise through the building envelope. Standard double glazing achieves a reduction of approximately 30–35 dB relative to single glazing, with further gains from triple glazing and improved frame sealing (??).
IEQ outgoing edges	
IEQ → PH	Adverse indoor environmental conditions — including dampness, mould, cold temperatures, and elevated concentrations of indoor pollutants — cause respiratory disease, cardiovascular strain, and excess winter mortality. Meta-analyses of controlled and observational evidence report odds ratios of 1.34–1.75 for respiratory outcomes associated with damp and mouldy housing (???).
Social and Psychological Mediator edges	
Accessibility → PH	Physical access to healthcare determines the timeliness and continuity of treatment for acute and chronic physical illness, operating through deferred diagnosis, missed appointments, and discontinuous management of long-term conditions. A scoping review of distance and travel time to health services in rural and remote areas finds that greater distance and travel time are associated with delayed presentation, lower service utilisation, and worse health outcomes, with effects concentrated among elderly and chronically ill populations (?). UK health-service inequalities research documents the same area-level access burden (?). This pathway is mechanistically distinct from *Accessibility* → *MH* (Table 7): the present edge captures the consequences of constrained physical-healthcare access for physical morbidity and mortality, rather than the psychological effects of restricted participation. Because *PH* → *MH* already exists, this edge also opens an indirect *Accessibility* → *PH* → *MH* sub-path running parallel to the direct *Accessibility* → *MH* channel.
Social_Capital → Isolation	Strong community networks reduce social isolation by facilitating incidental interaction, reinforcing trust, and providing a sense of belonging. Evidence from cross-sectional and longitudinal studies consistently identifies higher social capital as protective against loneliness and social withdrawal (??).

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Edge	Rationale and supporting evidence
Social_Capital → Substance_Abuse	Community cohesion protects against substance misuse through the enforcement of social norms and the provision of informal support networks. A systematic review of alcohol interventions finds that higher social capital is associated with significantly reduced binge drinking (adjusted OR 0.38), with complementary evidence on the broader protective role of community ties (??).
Digital_Capital → Isolation	Digital exclusion limits access to online social interaction and public services, increasing the risk of loneliness. An analysis of the English Longitudinal Study of Ageing (ELSA) finds that digital exclusion is associated with higher odds of loneliness (OR 1.30), with comparable evidence from additional cohort studies (??).
Substance_Abuse → PH	Alcohol and drug misuse cause liver disease, cardiovascular damage, cancer, and premature mortality through direct physiological mechanisms. A Global Burden of Disease analysis estimates that 2.8 million deaths globally were attributable to alcohol consumption in 2016 (?).
Demographics outgoing edges	
Demographics → Housing	Ethnicity, age, and household composition influence housing quality through differential access to housing markets and documented discrimination in allocation and lending. Pakistani and Black Other households experience overcrowding rates exceeding 20%, reflecting structurally unequal outcomes in housing access (??).
Demographics → Overcrowding	Household composition and ethnicity are among the strongest determinants of overcrowding. 2021 Census data show that Black households faced an overcrowding rate of 16.1%, compared with 2.5% for White households, illustrating the magnitude of demographic disparities in housing density (??).
Demographics → Economic_Capital	Gender, ethnicity, and age systematically shape earnings through labour market discrimination and structural barriers to employment and progression. Bangladeshi employees face the largest documented negative ethnicity pay gap (−16%), with parallel penalties associated with gender and age (???).
Demographics → Education	Socioeconomic background, gender, and ethnicity are significant predictors of educational attainment. At age 16, the socioeconomic achievement gap is approximately twice the size of the largest ethnic attainment gap and six times the gender gap, underscoring the dominance of socioeconomic origin in shaping educational outcomes (??).

Table 16: Edges considered but excluded from the DAG, with the prima facie case for inclusion and the reason for omission.

Excluded edge	**Prima facie case and reason for exclusion**
Education → HEE	Higher educational attainment might plausibly increase retrofit uptake through improved energy literacy, awareness of subsidy programmes, and capacity to evaluate long-term returns on efficiency investments. However, empirical analysis of British household survey data finds that a BA-degree indicator is not a statistically significant predictor of retrofit investment once income and dwelling characteristics are controlled for (**?**). Cross-national studies reporting positive education-retrofit associations attribute them predominantly to education's correlation with income rather than to an independent information channel. We therefore model the effect of education on HEE as operating entirely through *Education* → *Economic_ Capital* → *HEE*, and omit the direct edge.
Demographics → HEE	Fuel poverty incidence and EPC distribution differ systematically by ethnicity, age, and household composition in England, making a direct demographic effect on HEE intuitive. Direct empirical tests of this question, however, consistently find that demographic variables' explanatory power over HEE operates predominantly through economic and housing mediators already represented in the DAG. A recent machine learning analysis at LSOA level in England and Wales using 2021 Census and EPC data — the scale and setting of the present analysis — finds that socio-demographic variables collectively explain approximately one-third of the variance in HEE, but that the dominant predictors are the National Statistics Socio-Economic Classification, household deprivation, and median age, while ethnicity and religious affiliation exert comparatively modest influence (**?**). The leading predictors in that analysis are direct proxies for *Economic_ Capital* rather than for *Demographics* as constructed in this DAG, and the analysis does not condition on dwelling characteristics, which separate work has consistently found to be the dominant determinant of HEE (**????**). These findings are consistent with earlier evidence that socio-demographic characteristics are substantially less important than dwelling-related characteristics in predicting retrofit uptake in English households once income is controlled for. Demographics' effect on HEE is therefore adequately captured through the established indirect pathways *Demographics* → *Housing_ Characteristics* → *HEE*, *Demographics* → *Economic_ Capital* → *HEE*, and *Demographics* → *Tenure* → *HEE*, and a direct edge would risk double-counting effects already transmitted through these mediators. Fuel poverty, a joint function of income and HEE (**?**), is not equivalent to HEE and is treated accordingly.

Table 16 continued from previous page

Excluded edge	Prima facie case and reason for exclusion
Housing_Characteristics → MH	A direct effect of physical dwelling form on mental health is suggested by prospective evidence that persistent poor housing conditions predict psychological distress over multiyear horizons (**??**). In the present DAG, however, *Housing_Characteristics* is defined as physical dwelling form (age, construction era, type, materials), while housing *quality* — damp, mould, cold, and indoor pollutants, which are the proximal drivers of the mental health effects these studies document — is represented by the *IEQ* node. The mental health consequences of physical housing are therefore transmitted through *Housing_Characteristics* → *IEQ* → *MH*, with additional pathways via *Overcrowding* and *Isolation*. A direct edge would double-count the IEQ-mediated effect that constitutes the dominant evidence base in the housing-mental-health literature.
Neighbourhood_Stress → Green_Space	Fear of crime and neighbourhood disorder reduce residents' behavioural engagement with local parks and green infrastructure, particularly among women and ethnic minorities (**??**). In the present DAG, however, *Green_Space* is defined as the area-level provision and quality of parks, vegetation, and blue space, measured at LSOA level through land-cover and density indicators rather than use or engagement data. The behavioural effect of neighbourhood stress on green-space engagement is conceptually distinct from provision, and its mental health consequences are captured through two existing pathways: *Neighbourhood_Stress* → *Isolation* → *MH* (reduced social interaction) and the direct *Neighbourhood_Stress* → *MH* channel (chronic stress). A direct edge to *Green_Space* would require redefining that node to include use, breaking consistency with Tables 7 and 12.

Table 16 above lists edges that are genuinely absent from the DAG. Several other edges are present and committed to a single orientation but describe relationships whose direction expert review judged arguable; these are termed *orientation-uncertain* throughout, and for them the question is not whether to draw the edge but whether its orientation affects identification. The term is distinct from a *bidirected* edge, which is the $\leftrightarrow$ mark used below and in the partial ancestral graphs of Section 4.3 to denote an unmeasured common cause, and from a *reciprocal* relationship, meaning genuine mutual causation, which acyclicity forbids. Table 17 reports a joint orientation-sensitivity analysis of the total effect of HEE on MH over all $3^8 = 6561$ configurations in which each of eight candidate edges is held at its committed orientation, reversed, or replaced by a latent common cause (a bidirected $\leftrightarrow$ edge). The committed (all-original) configuration returns the compact minimal adjustment set $\mathcal{A} = \{\text{Climate, Economic_Capital, Housing_Characteristics, Tenure, Urbanicity}\}$, vindicating the DAG as drawn: of the 6561 configurations, 73.7% are cyclic, 18.9% are identifiable, and only 7.4% are non-identifiable. Non-identifiability has a single gateway — a latent HEE $\leftrightarrow$ Economic_Capital common cause is present in all 486 of 486 non-identifiable configurations — with exactly two minimal (Hamming-distance-2) breakers: HEE $\leftrightarrow$ EC paired with EC $\leftrightarrow$ MH, and HEE $\leftrightarrow$ EC paired with EC $\leftrightarrow$ PH. No outcome-side reorientation breaks identification. Because reversal is itself cyclic for half the candidates, the substantively meaningful robustness operation is the latent common cause ($A \leftrightarrow B$), not arrow-flipping. Three entries previously listed as "excluded" reverse directions (HEE $\rightarrow$ Economic_Capital, Economic_Capital $\rightarrow$ Education, and MH $\rightarrow$ PH) are superseded by this analysis and have been moved here, with their original rationale preserved in the Resolution column.

Table 17: Orientation-uncertain edges and the joint orientation-sensitivity verdict for the total effect of HEE on MH. Of the $3^8 = 6561$ configurations, 73.7% are cyclic, 18.9% identifiable and 7.4% non-identifiable; a latent common cause of HEE and *Economic_Capital* is present in all 486 non-identifiable configurations. The DAG topology is unchanged; all eight committed orientations are retained. Source: full 3^8 enumeration, `bidirectionality_sensitivity.md`.

Edge	**Committed orientation**	**Class**	**Effect on identification**	**Resolution**
HEE — Economic_Capital	Economic_Capital → HEE	Confounder-side	Most load-bearing edge and the *sole* gateway to non-identifiability: a latent HEE ↔ Economic_Capital common cause is present in all 486 of 486 non-identifiable configurations and forms both minimal breakers when paired with a latent EC–outcome leg.	Resolved by temporal ordering. Economic Capital is a baseline / pre-treatment confounder, measured before the HEE exposure contrast, which licenses the acyclic *Economic_Capital* → *HEE* encoding; the latent variant corresponds to the case where that temporal separation fails. The reverse *HEE* → *Economic_Capital* edge is not drawn: energy efficiency improvements reduce household energy bills, which at the household level represents an increase in disposable income. Including this edge would make HEE a cause of downstream economic capital and would substantially complicate identification. At the LSOA area level, however, the bill-saving effect of HEE is already represented within the DAG through the affordability channel to mental health (*HEE* → *IEQ* captures thermal comfort; the financial stress relief operating through reduced fuel poverty is captured in the HEE → MH mechanism documented in **?** and **?**). Adding a direct *HEE* → *Economic_Capital* edge would conflate the exposure of interest with one of its downstream consequences, rendering the target estimand ambiguous. The edge is omitted, and the affordability mechanism is acknowledged as part of the HEE–MH pathway rather than as a modifier of the confounder set.

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Edge	Committed orientation	Class	Effect on identification	Resolution
Economic_Capital — Education	Education → Economic_Capital	Confounder-side	Inert in isolation but jointly load-bearing: pivotal in 210 of the joint configurations, though it does not alter $\mathcal{A}$ in the committed configuration.	Resolved by temporal ordering (baseline measurement). Education and Economic Capital are ascertained at baseline, before the HEE exposure contrast, so the acyclic *Education* → *Economic_Capital* direction encodes a feedback system observed at fixed times. The reverse *Economic_Capital* → *Education* edge is not drawn: parental economic resources shape children's educational opportunities through school quality, tutoring, and sustained enrolment; this intergenerational mechanism is well-documented (**?**). Including this edge would, however, introduce a cycle with the existing *Education* → *Economic_Capital* edge, which captures the adult labour-market returns to schooling. DAG formalism requires a directional choice. The *Education* → *Economic_Capital* direction is retained because (i) at any single cross-section at LSOA level, the dominant data-generating process for working-age earnings is the labour-market-returns mechanism; (ii) the intergenerational reverse-direction mechanism operates on timescales outside the temporal scope of the present analysis; and (iii) the latent node *U_AreaHistory* already absorbs part of the area-level intergenerational persistence that would otherwise require a *Economic_Capital* → *Education* edge (**?**). The reciprocal nature of the underlying relationship is an acknowledged limitation rather than a modelling error.

Table 17 continued from previous page

Edge	Committed orientation	Class	Effect on identification	Resolution
Economic_Capital — Policy_Scheme_Access	Economic_Capital → Policy_Scheme_Access	Confounder-side	Load-bearing: reversal to *Policy_Scheme_Access* → *Economic_Capital* forces Policy_Scheme_Access into the adjustment set.	Resolved by defended orientation plus a bounded failure mode. The committed *Economic_Capital* → *Policy_Scheme_Access* orientation is defended because ECO, GBIS and the Warm Home Discount are means-tested and area-targeted, so area economic capital determines scheme access. Under *Economic_Capital* → *Policy_Scheme_Access*, Policy_Scheme_Access is a descendant of Economic_Capital and is not required in $\mathcal{A}$. Under the reversed orientation *Policy_Scheme_Access* → *Economic_Capital* it would instead be a required but *unmeasured* confounder — stated here as a named, bounded failure mode rather than silently dropped.
Economic_Capital — MH	Economic_Capital → MH	Confounder-side	A latent Economic_Capital ↔ MH common cause is one of the two minimal (Hamming-2) non-identifiability legs, in conjunction with a latent HEE ↔ Economic_Capital link.	Committed *Economic_Capital* → *MH* retained; Economic Capital is measured and adjusted within $\mathcal{A}$. Non-identifiability arises only if this association is *additionally* treated as an unmeasured common cause jointly with a latent HEE ↔ Economic_Capital link — i.e. only when Economic Capital is given an unmeasured common cause with both the exposure and the outcome.

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Edge	Committed orientation	Class	Effect on identification	Resolution
PH — MH	PH → MH	Outcome-side	Indifferent for identification: neither the reported set $\mathcal{A}$ nor identifiability changes under reversal or a latent variant; only non-reported alternative minimal sets shift. Listed for transparency.	Committed *PH* → *MH* retained. Common mental disorders increase physical morbidity through neuroendocrine, inflammatory, and behavioural pathways; mediation analyses using UK data find significant effects in both the physical-to-mental and mental-to-physical directions (**??**). The relationship is genuinely reciprocal, but the DAG framework requires a single directional choice. We retain *PH* → *MH* and omit *MH* → *PH* because, for the target estimand (the effect of HEE on MH), the *PH* → *MH* direction is what creates backdoor paths through physical health and therefore what matters for confounder identification. Prioritising the direction that threatens identification is the more conservative modelling choice; the reverse pathway, while real, would introduce a cycle without altering the adjustment set for our exposure-outcome pair. The implicit feedback loop is an acknowledged limitation of a cross-sectional DAG applied to a substantively dynamic system.
Substance_Abuse — MH	Substance_Abuse → MH	Outcome-side	Fully robust: never pivotal among valid graphs (0 of 2187 contexts). Orientation never changes any adjustment set; reversal only ever induces a cycle. Listed for transparency.	Committed *Substance_Abuse* → *MH* retained; its orientation is immaterial to identification of the HEE–MH effect, and the only consequence of reversal is an invalid (cyclic) graph.

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Edge	Committed orientation	Class	Effect on identification	Resolution
Isolation — MH	Isolation → MH	Outcome-side	Indifferent for identification: the reported set $\mathcal{A}$ and identifiability are unaffected; only non-reported alternative minimal sets shift. Listed for transparency.	Committed *Isolation* → *MH* retained; the orientation does not affect the reported adjustment set $\mathcal{A}$ or identifiability of the HEE–MH effect.
Economic_Capital — PH	Economic_Capital → PH	Confounder-side	A latent Economic_Capital ↔ PH common cause is the other minimal (Hamming-2) non-identifiability leg, in conjunction with a latent HEE ↔ Economic_Capital link.	Committed *Economic_Capital* → *PH* retained; both Economic Capital and PH are handled within the graph. Non-identifiability arises only if this association is *additionally* treated as an unmeasured common cause jointly with a latent HEE ↔ Economic_Capital link.

B. Causal-Discovery Comparison Tables

Table 18
Agreement between the PC-stable / FCI discovered graphs and the induced measured subgraph of the expert DAG (16 nodes, 40 directed edges; MH = QOF depression). Skeleton precision/recall are for undirected adjacency; SHD counts missing, extra, and orientation-mismatched edges.

Configuration	Edges	Skel. P	Skel. R	SHD	V-struct R
PC α=0.01, no tiers	45	0.42	0.47	62	0.01
PC α=0.01, tiers	45	0.42	0.47	50	0.14
PC α=0.005, no tiers	43	0.42	0.45	61	0.01
PC α=0.005, tiers	43	0.42	0.45	50	0.13
PC α=0.001, no tiers	44	0.41	0.45	62	0.01
PC α=0.001, tiers	44	0.41	0.45	51	0.13
FCI α=0.01	39	0.41	0.40	61	0.11
FCI α=0.005	37	0.43	0.40	59	0.09
FCI α=0.001	37	0.43	0.40	59	0.09

Table 19
Edge-level disagreement between the expert induced subgraph and the PC-stable CPDAG (α=0.01, no tiers, MH = QOF depression). (a) expert adjacencies not recovered; (b) recovered adjacencies absent from the expert DAG. Candidate explanations are given in the accompanying markdown.

(a) In expert subgraph, not recovered	
Accessibility	Mental Health
Air Pollution	Green Space
Air Pollution	Physical Health
Climate	Housing Characteristics
Climate	Home Energy Efficiency
Demographics	Economic Capital
Demographics	Education
Demographics	Housing Characteristics
Demographics	Mental Health
Economic Capital	Housing Characteristics
Economic Capital	Home Energy Efficiency
Economic Capital	Mental Health
Economic Capital	Urbanicity
Education	Urbanicity
Fuel Poverty	Mental Health
Green Space	Mental Health
Green Space	Physical Health
Housing Characteristics	Physical Health
Home Energy Efficiency	Urbanicity
Mental Health	Neighbourhood Stress
Mental Health	Tenure
(b) Recovered, absent from expert subgraph	
Accessibility	Air Pollution
Accessibility	Green Space
Accessibility	Home Energy Efficiency
Accessibility	Neighbourhood Stress
Accessibility	Overcrowding
Accessibility	Urbanicity
Air Pollution	Climate
Air Pollution	Demographics
Air Pollution	Neighbourhood Stress
Air Pollution	Overcrowding
Air Pollution	Urbanicity
Climate	Green Space
Climate	Urbanicity
Demographics	Green Space
Demographics	Neighbourhood Stress
Demographics	Tenure
Demographics	Urbanicity
Economic Capital	Neighbourhood Stress
Education	Fuel Poverty
Fuel Poverty	Housing Characteristics
Green Space	Home Energy Efficiency
Green Space	Neighbourhood Stress
Housing Characteristics	Overcrowding
Housing Characteristics	Urbanicity
Neighbourhood Stress	Tenure
Tenure	Urbanicity